

ECE 253

Guest lecture on Color
Pamela Cosman

October 8, 2024

Color Image Processing

- **Color representation**

- Why do we use 3 color planes? Why not 2 or 4?
- Why R, G, B?
- Can we display any color on our screens?
- Alphabet soup of other representation systems: XYZ, HSI, YIQ, CMK, Lab, etc.

- **Color image processing**

- Which color representation system to use for image processing?
- Do we have to modify a grayscale algorithm to be suitable for color images?

Perceptual Descriptors of Color

When people asked to describe their perception of a color sensation, they generally describe in terms of:

- **Intensity / Brightness:** Is there more or less light?



- **Saturation:** Is the color saturated (deep, pure, vivid, rich, strong) or unsaturated (washed out, pale, pastel, weak)?



- **Hue:** Is the color similar to red, blue, yellow, etc?



Hue and saturation together called *chromaticity*

Achromatic light is void of color, has intensity characteristic only

Newton: Separation and mixtures of wavelengths



<https://www.biography.com/news/how-isaac-newton-changed-our-world>

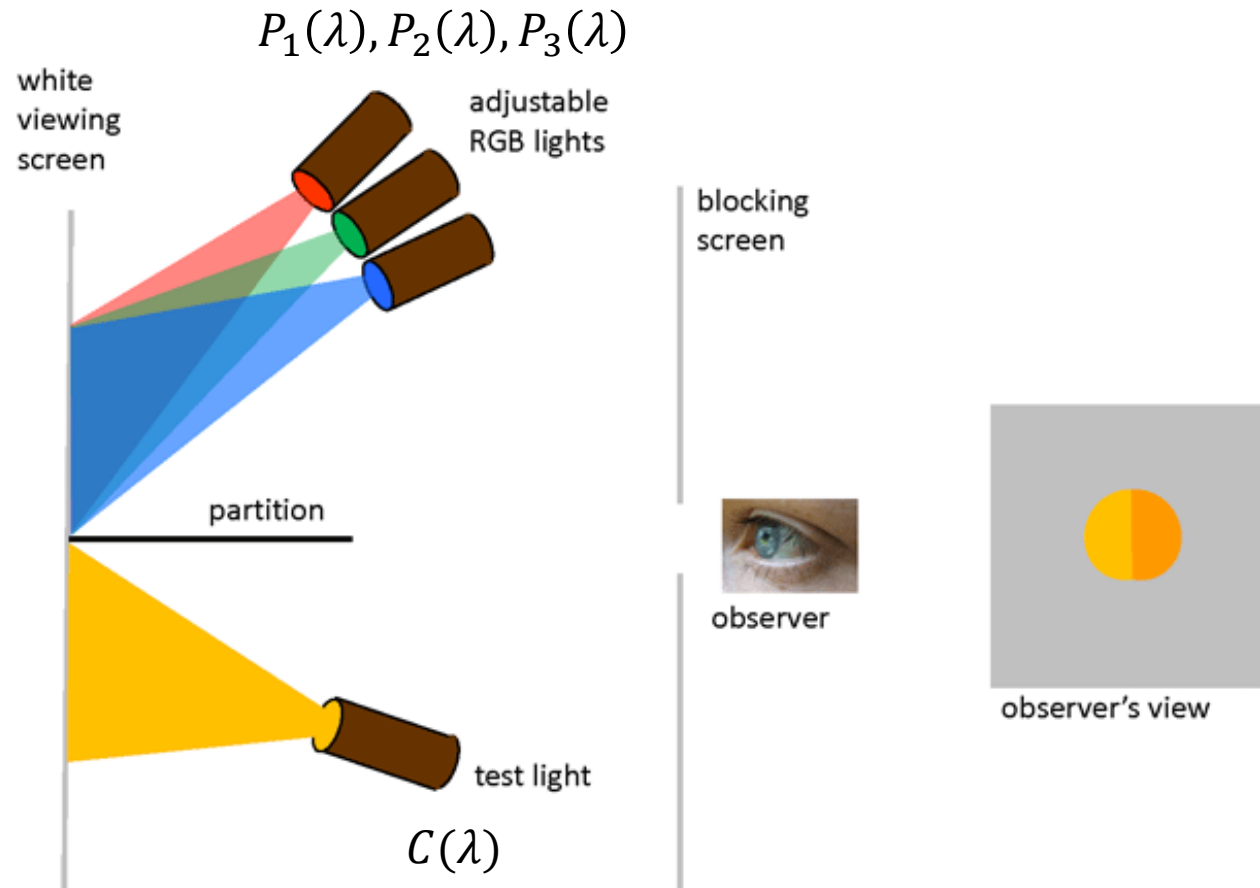
380-450nm	425-450?	450-495	495-570	570-590	590-620	620-750
violet	indigo?	blue	green	yellow	orange	red

ROYGBIV

Newton started with 5 colors, went to 7, because of # days in week, notes in musical scale...

Simple observations:
Red + Yellow looks like Orange

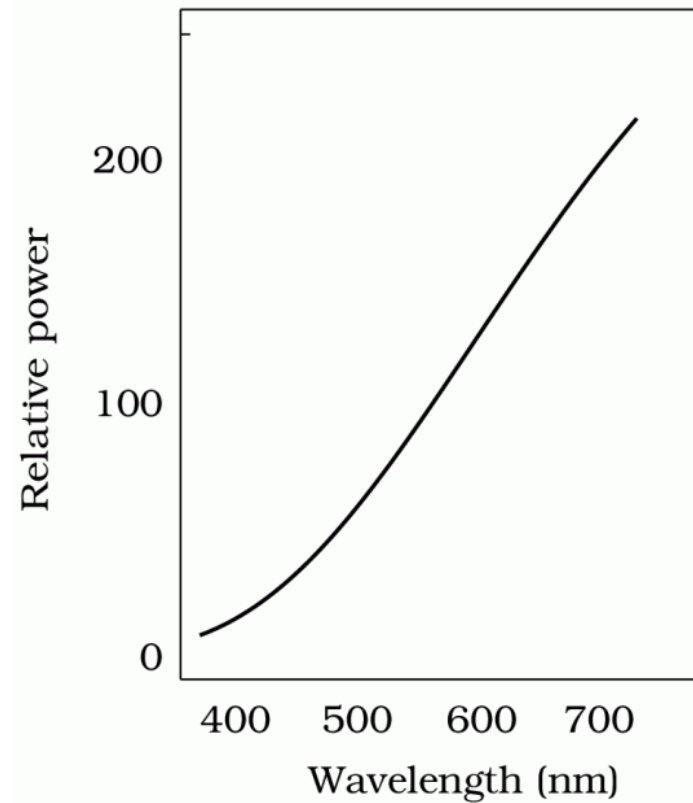
Color Matching Experiment



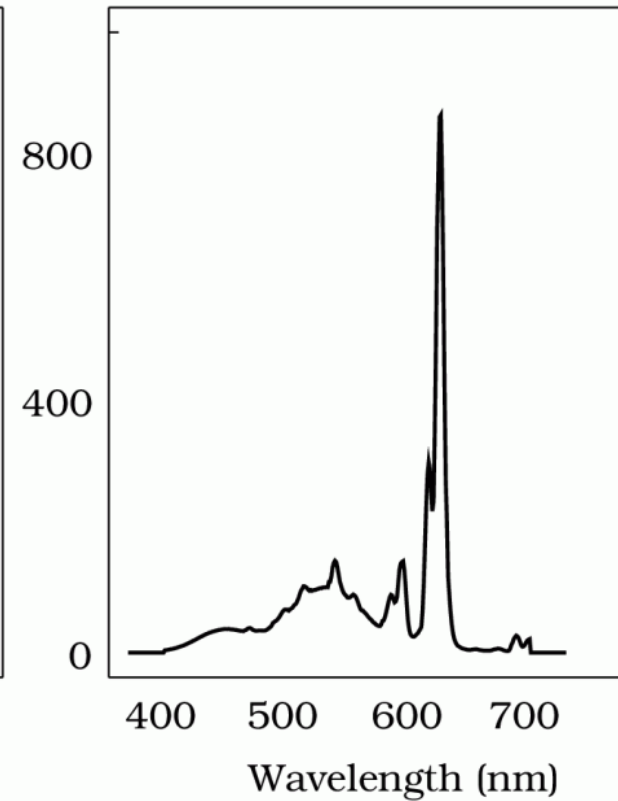
- **Test light:** $C(\lambda)$ has any spectral distribution. Observer cannot change it.
- **Primary lights:** $P_1(\lambda)$, $P_2(\lambda)$, $P_3(\lambda)$ have any* spectral distributions. Observer can adjust the overall gain for each one separately, but cannot change the shape of the distribution
- **Goal:** Adjust the intensity factors until 2 sides look identical
- **Results:** For any* test light, can get a match! Radically different spectra on the 2 sides, look the same.
- The match is *unique*

Example of metamers

Metamer: A color that appears to the eye to be identical to another color but which in fact has a different spectral composition.



Sunlight



Match on a TV

Normalization to a reference white

How do we report the results of this experiment?

First choose some reference white light: $W(\lambda)$

Do the experiment to match it

Let $A_1(W), A_2(W), A_3(W)$ represent the positions of the 3 knobs for matching $W(\lambda)$

$$A_1(W)P_1(\lambda) + A_2(W)P_2(\lambda) + A_3(W)P_3(\lambda) \leftrightarrow W(\lambda)$$



This symbol means "matches"
or "is a visual match for"

$\leftrightarrow$ does NOT mean =

The spectrum on the left is a metamer of the spectrum on the right

Tristimulus Values

In the same way, for any test light $C(\lambda)$, we can write:

$$A_1(C)P_1(\lambda) + A_2(C)P_2(\lambda) + A_3(C)P_3(\lambda) \leftrightarrow C(\lambda)$$

where $A_1(C), A_2(C), A_3(C)$ represent the positions of the 3 knobs for matching $C(\lambda)$

We normalize using the values for the reference white:

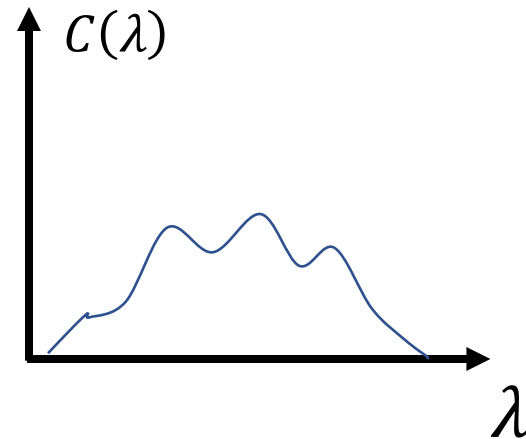
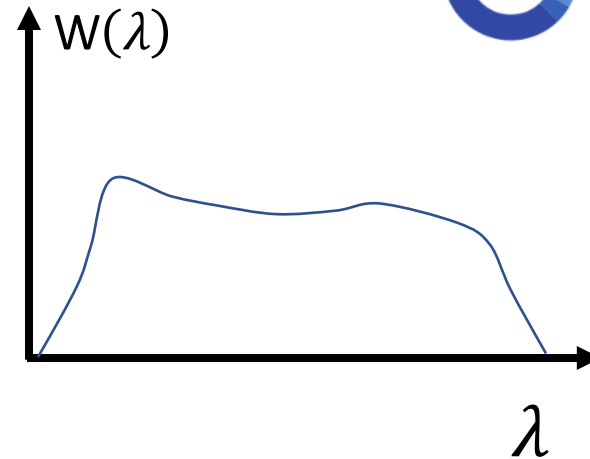
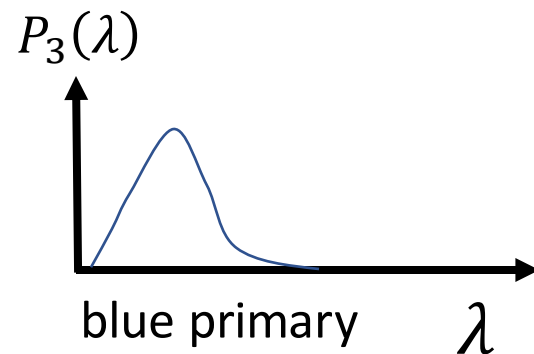
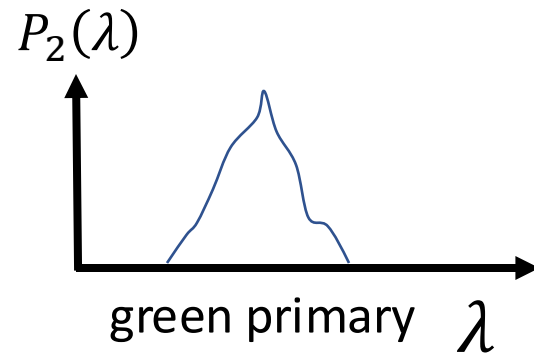
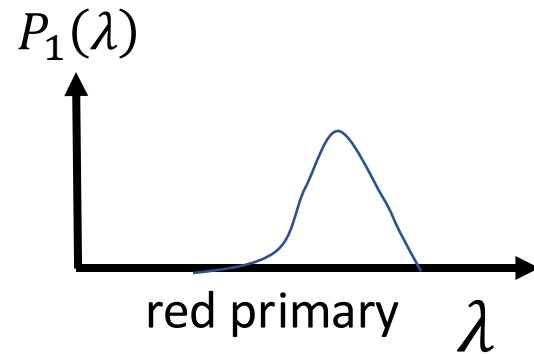
$$T_1(C) = \frac{A_1(C)}{A_1(W)} \quad T_2(C) = \frac{A_2(C)}{A_2(W)} \quad T_3(C) = \frac{A_3(C)}{A_3(W)}$$

$T_1(C), T_2(C), T_3(C)$ are called the *tristimulus values* for the color $C(\lambda)$

$\leftrightarrow$ means “visual match”
It does NOT mean =

Q: What are the
tristimulus values of
the reference white?

Many different
reference whites



Units?

These are spectral power distributions

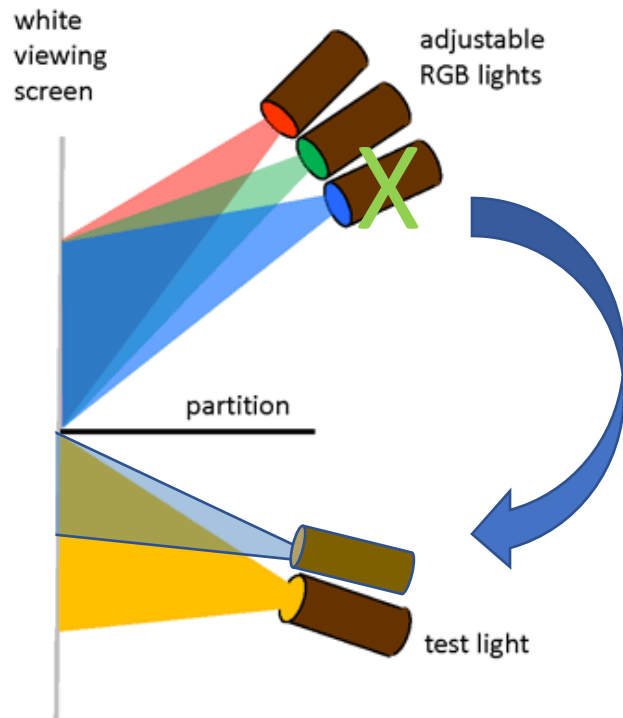
x-axis unit is nm

y-axis generally represents power per unit wavelength (Watts/nm)

- Watts $\rightarrow$ Watts/sr, Watts/m²

Why: “For any* test light, can get a match” ?

May need to unplug a primary, move it to the other side, add it to the test light



This counts as a negative tristimulus value

In fact, might need to unplug 2 primaries and move them both to the other side

If you allow unplugging & moving, can match any test light!

Why: “ $P_1(\lambda), P_2(\lambda), P_3(\lambda)$ have any* spectral distributions”

Need linearly independent primaries

You can't have:

$$aP_1(\lambda) + bP_2(\lambda) \leftrightarrow cP_3(\lambda)$$

If you can use two of the primaries (with some weight) to make a match to the third, they are called linearly dependent

Simplify notation:

$$T_1(C)P_1(\lambda) + T_2(C)P_2(\lambda) + T_3(C)P_3(\lambda) \leftrightarrow C(\lambda)$$

$$T_1P_1 + T_2P_2 + T_3P_3 \leftrightarrow C(\lambda)$$

It's a Linear System!

Satisfies homogeneity:

$$\text{If } T_1P_1 + T_2P_2 + T_3P_3 \leftrightarrow C(\lambda)$$

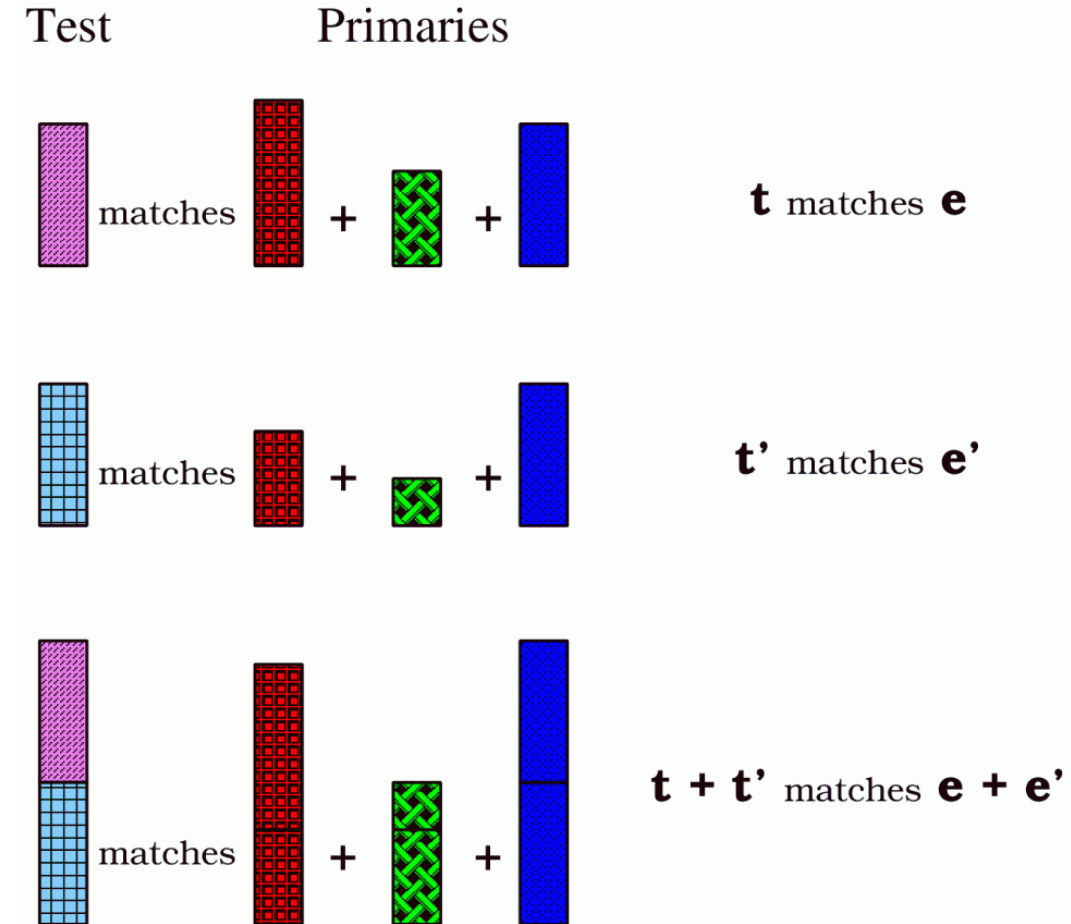
$$\text{then } aT_1P_1 + aT_2P_2 + aT_3P_3 \leftrightarrow aC(\lambda)$$

And superposition:

$$\text{If } T_1P_1 + T_2P_2 + T_3P_3 \leftrightarrow C(\lambda)$$

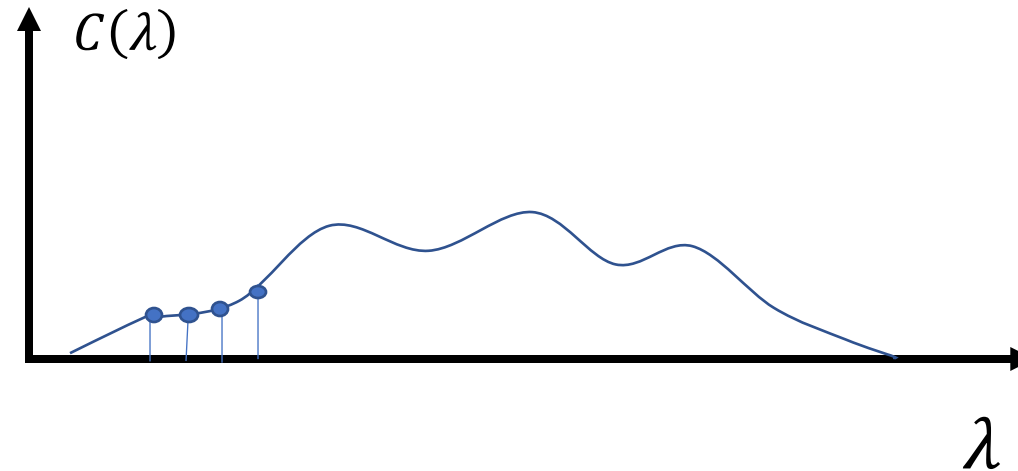
$$\text{and } T'_1P_1 + T'_2P_2 + T'_3P_3 \leftrightarrow C'(\lambda)$$

$$\text{then } (T_1 + T'_1)P_1 + (T_2 + T'_2)P_2 + (T_3 + T'_3)P_3 \leftrightarrow C(\lambda) + C'(\lambda)$$



Algebraic formulation of color

Arbitrary light
 $C(\lambda)$ sampled
 over visible range



$$C = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

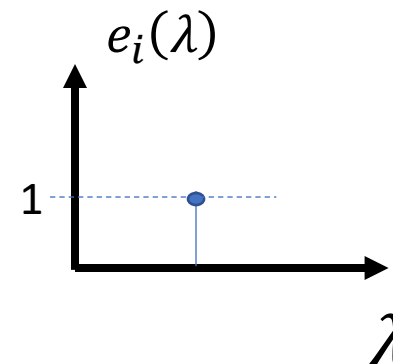
For example, sampled from 400nm to 700nm at intervals of 5nm

$$C = c_1 e_1 + c_2 e_2 + \cdots + c_n e_n$$

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$e_n = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$



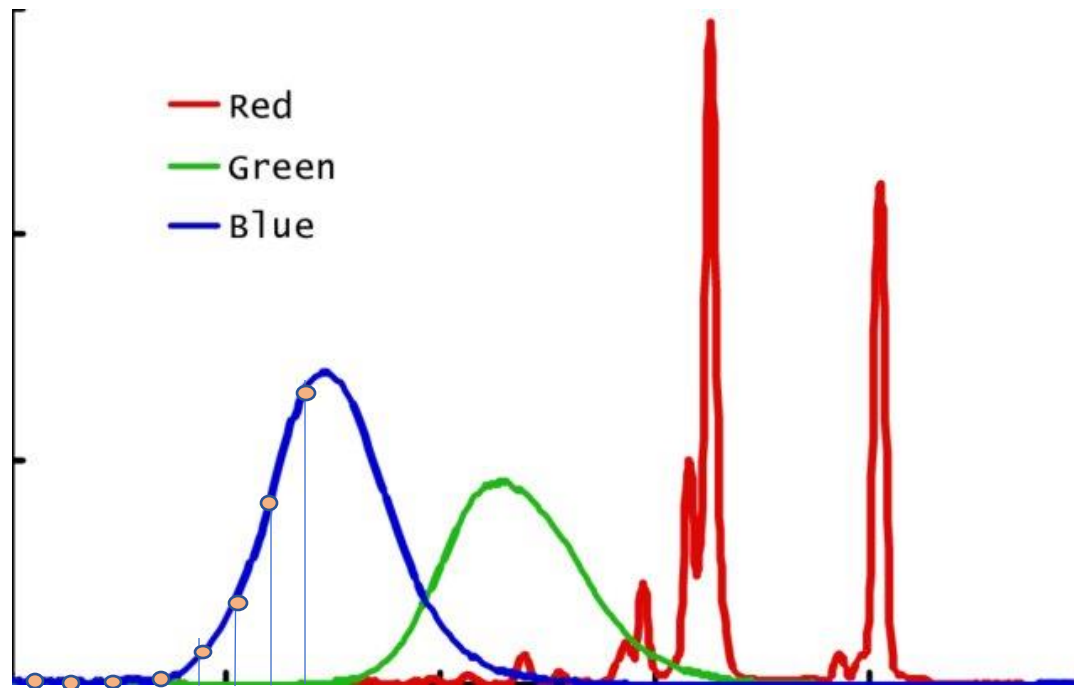
Monochromatic
 spectral (prism)
 colors

Algebraic formulation of color

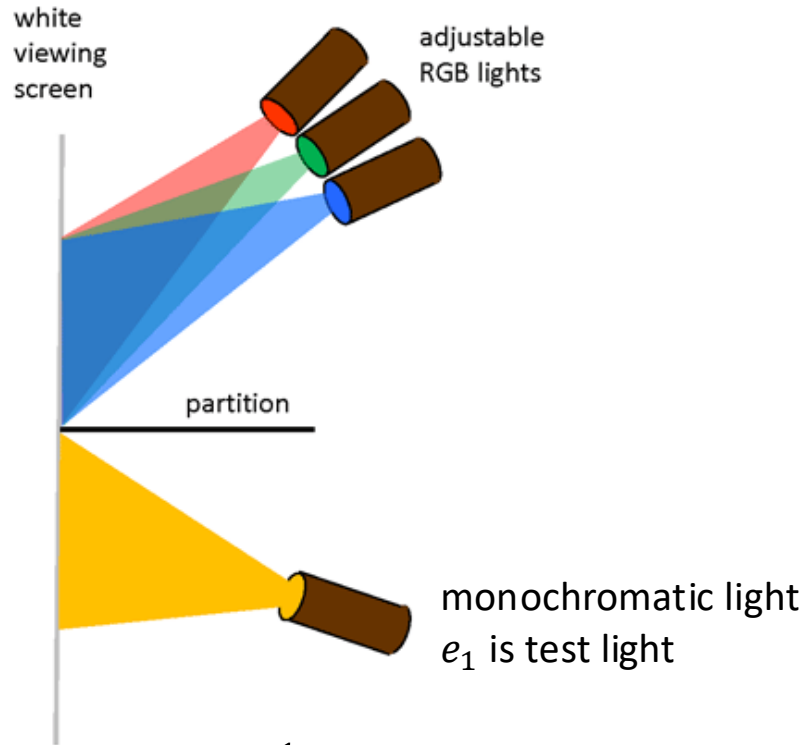
The primaries $P_1(\lambda), P_2(\lambda), P_3(\lambda)$ are also sampled:

$$P_1 = \begin{bmatrix} P_{1,1} \\ P_{1,2} \\ \vdots \\ P_{1,n} \end{bmatrix} \quad P_2 = \begin{bmatrix} P_{2,1} \\ P_{2,2} \\ \vdots \\ P_{2,n} \end{bmatrix} \quad P_3 = \begin{bmatrix} P_{3,1} \\ P_{3,2} \\ \vdots \\ P_{3,n} \end{bmatrix}$$

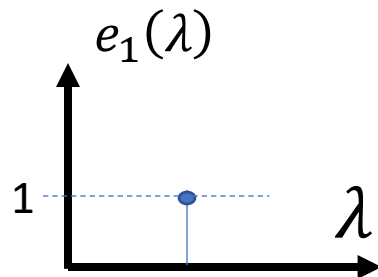
$$P = \begin{bmatrix} P_{1,1} & P_{2,1} & P_{3,1} \\ P_{1,2} & P_{2,2} & P_{3,2} \\ \vdots & \vdots & \vdots \\ P_{1,n} & P_{2,n} & P_{3,n} \end{bmatrix} \quad \begin{matrix} n \times 3 \\ \text{matrix} \end{matrix}$$



Match each of the monochromatic spectral lights



$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$



Use the color matching setup to match each one of the monochromatic spectral lights

$$m_{1,1} \begin{bmatrix} P_{1,1} \\ P_{1,2} \\ \vdots \\ P_{1,n} \end{bmatrix} + m_{2,1} \begin{bmatrix} P_{2,1} \\ P_{2,2} \\ \vdots \\ P_{2,n} \end{bmatrix} + m_{3,1} \begin{bmatrix} P_{3,1} \\ P_{3,2} \\ \vdots \\ P_{3,n} \end{bmatrix} \leftrightarrow e_1$$

$$\begin{bmatrix} | \\ a \\ | \end{bmatrix} \leftrightarrow \begin{bmatrix} | \\ b \\ | \end{bmatrix}$$

$$PM_1 \leftrightarrow e_1 \quad \text{where} \quad M_1 = \begin{bmatrix} m_{1,1} \\ m_{2,1} \\ m_{3,1} \end{bmatrix} \rightarrow \text{tristimulus values for } e_1$$

Do this for all of them:

$$PM_2 \leftrightarrow e_2 \quad PM_3 \leftrightarrow e_3 \text{ etc.}$$

Color Matching Functions

$$PM_1 \leftrightarrow e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad PM_2 \leftrightarrow e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad PM_n \leftrightarrow e_n = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

Stack up all those color matches into an array:

$$\underbrace{\begin{bmatrix} P_{1,1} & P_{2,1} & P_{3,1} \\ P_{1,2} & P_{2,2} & P_{3,2} \\ \vdots & \vdots & \vdots \\ P_{1,n} & P_{2,n} & P_{3,n} \end{bmatrix}}_P \times \underbrace{\begin{bmatrix} m_{1,1} & \dots & m_{1,i} & \dots & m_{1,n} \\ m_{2,1} & \dots & m_{2,i} & \dots & m_{2,n} \\ m_{3,1} & \dots & m_{3,i} & \dots & m_{3,n} \end{bmatrix}}_M$$

Color Matching Functions

$$PM \leftrightarrow I$$

$$\leftrightarrow \begin{bmatrix} 1 & & & & \\ & 1 & & & \\ & & \ddots & & \\ & & & 1 & \\ & & & & 1 \end{bmatrix}$$

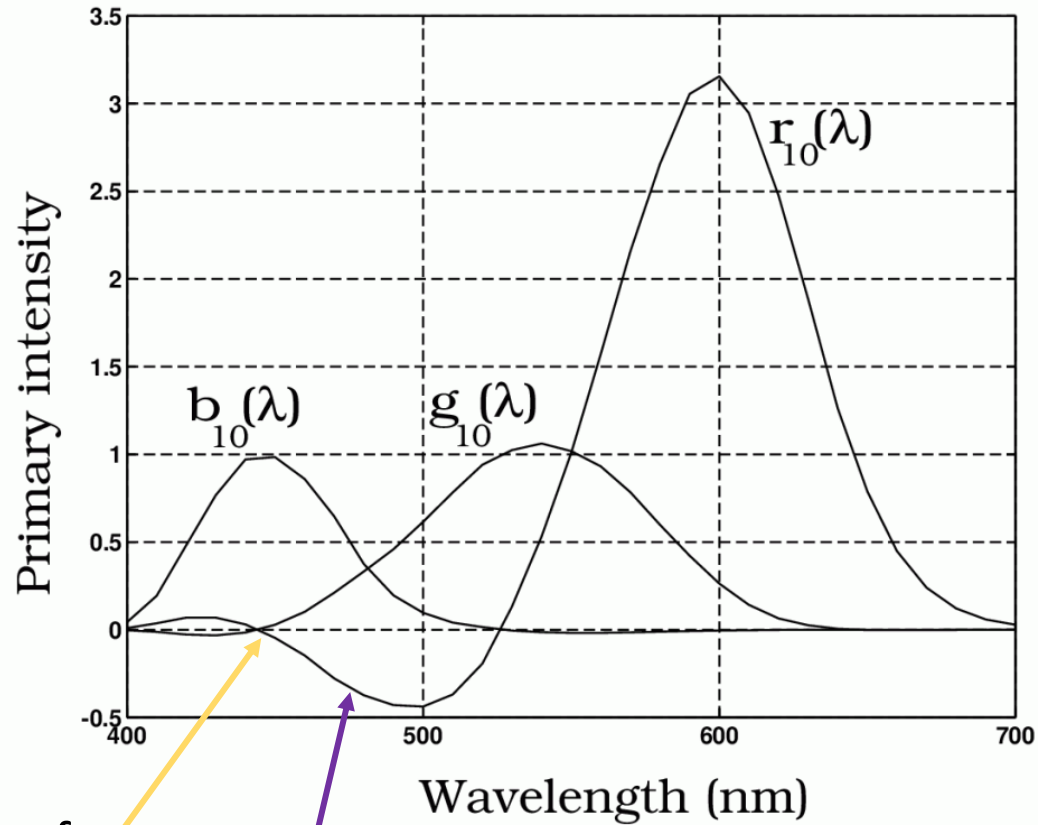
I

Abuse of notation:

Here the symbol $\leftrightarrow$ means that the first column on the left is a visual match for the first column on the right, etc.

Color Matching Functions

$$\underbrace{\begin{bmatrix} m_{1,1} & \dots & m_{1,i} & \dots & m_{1,n} \\ m_{2,1} & \dots & m_{2,i} & \dots & m_{2,n} \\ m_{3,1} & \dots & m_{3,i} & \dots & m_{3,n} \end{bmatrix}}_M$$



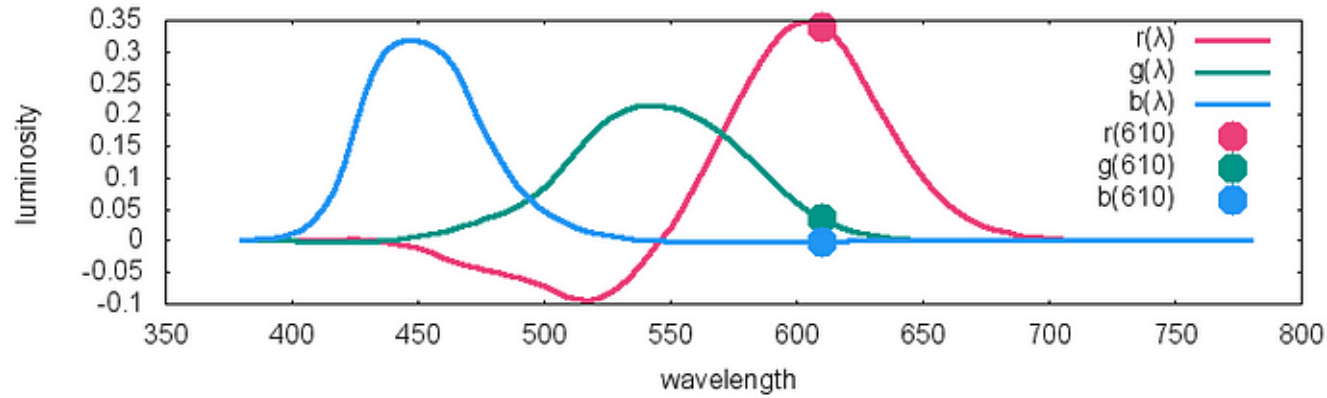
Primary lights:
645.2 nm,
526.3 nm, and
444.4 nm

Functions measured by
Stiles and Burch (1959)
using a 10 deg bipartite field
and primary lights at
wavelengths 645.2 nm,
526.3 nm, and 444.4 nm
with unit radiant power.

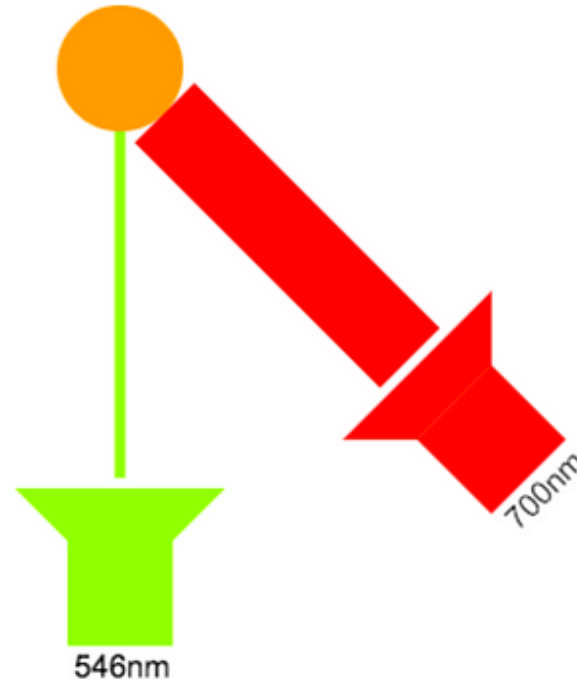
Two of
them are
zero here

Negative
values

Color matching:

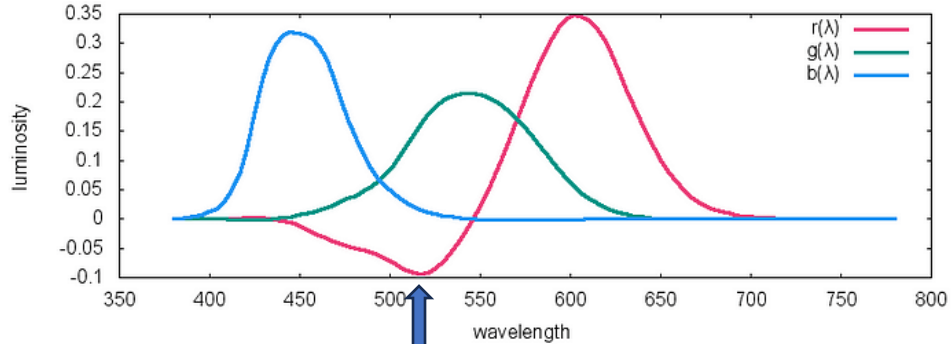


Matching the color of 610nm using 3 primaries

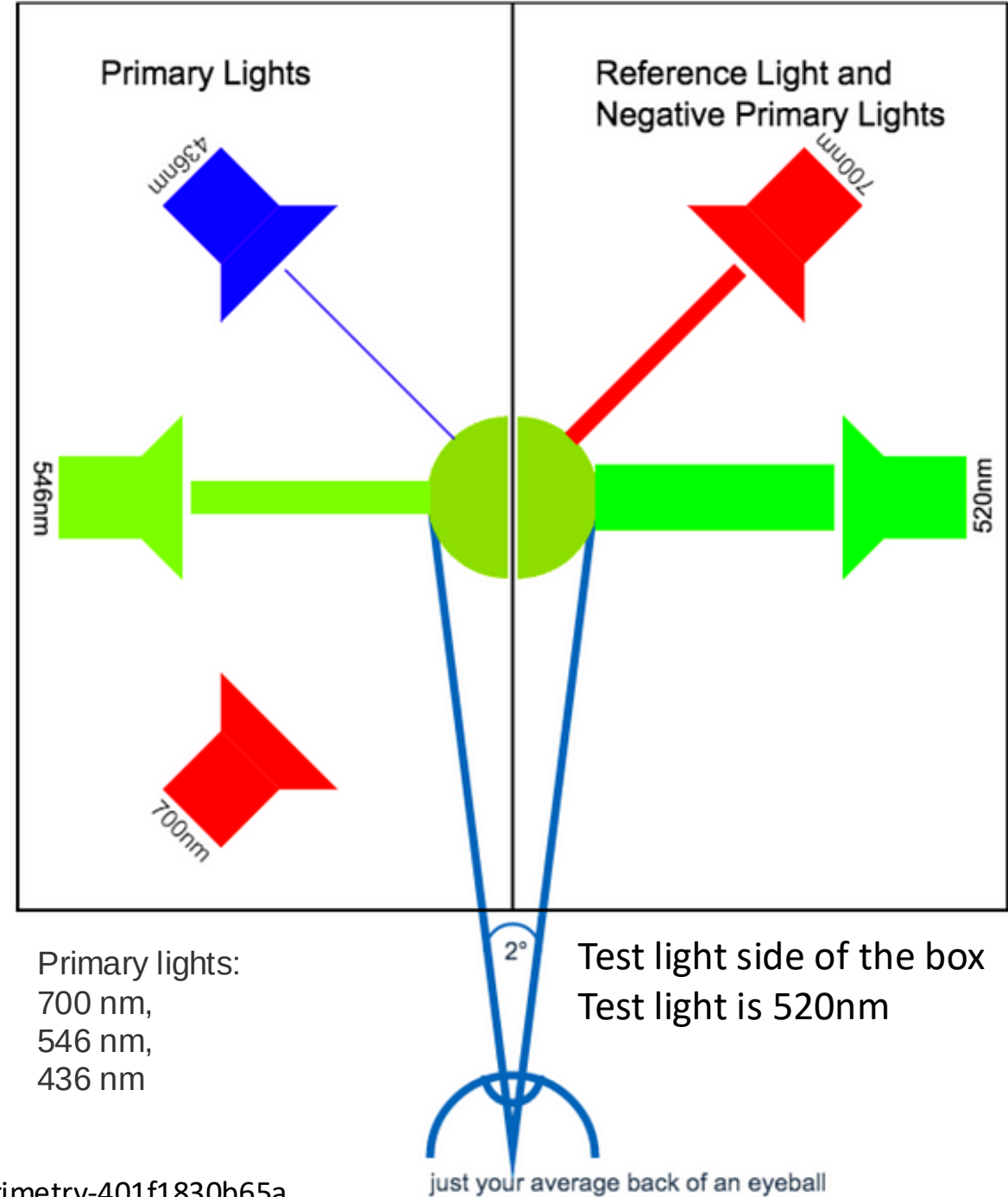


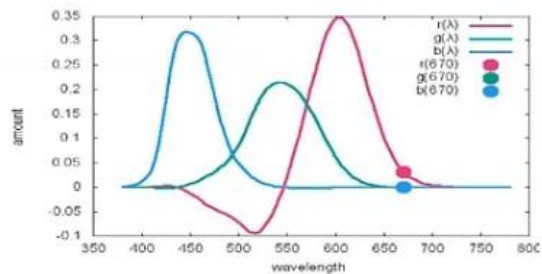
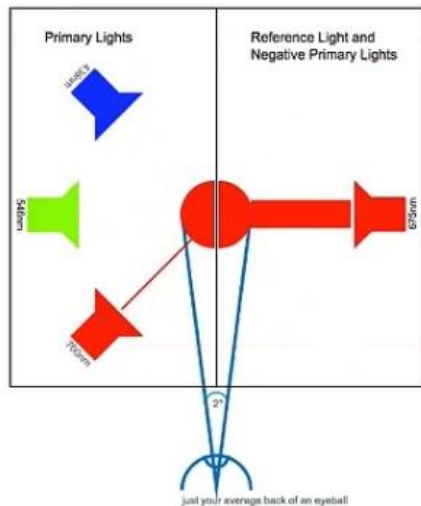
<https://medium.com/hipster-color-science/a-beginners-guide-to-colorimetry-401f1830b65a>

Color matching functions: Negative tristimulus values

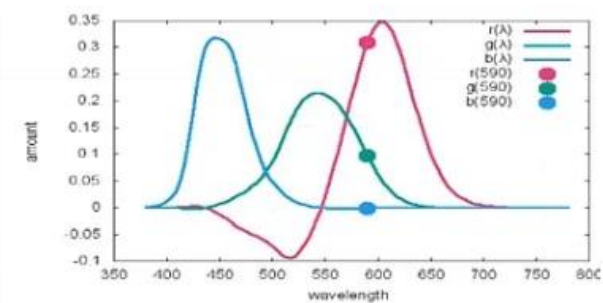
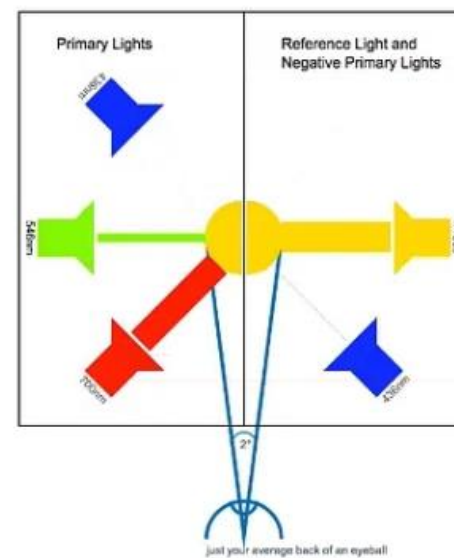
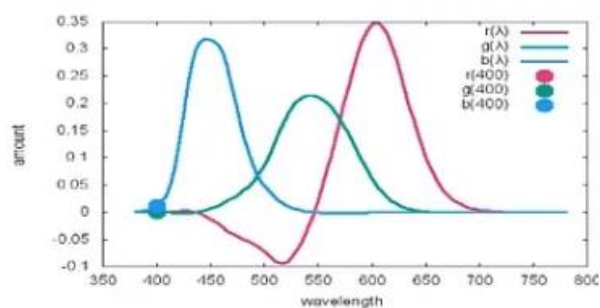
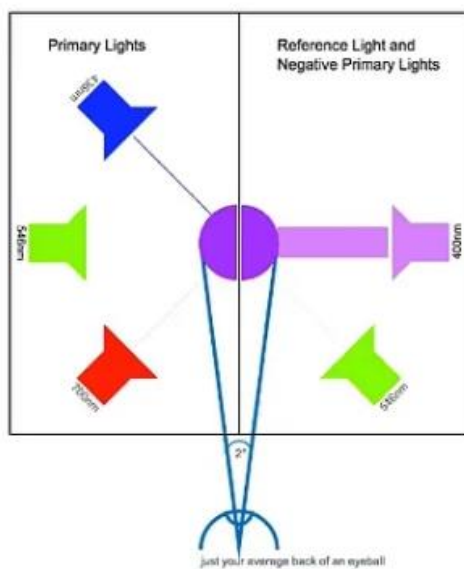
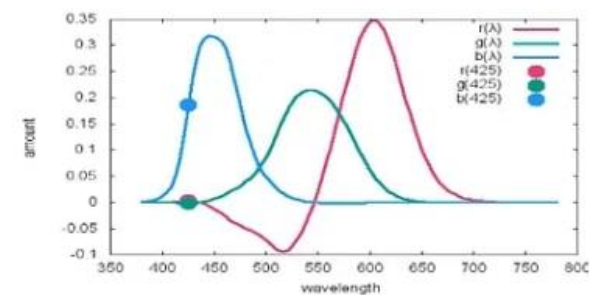
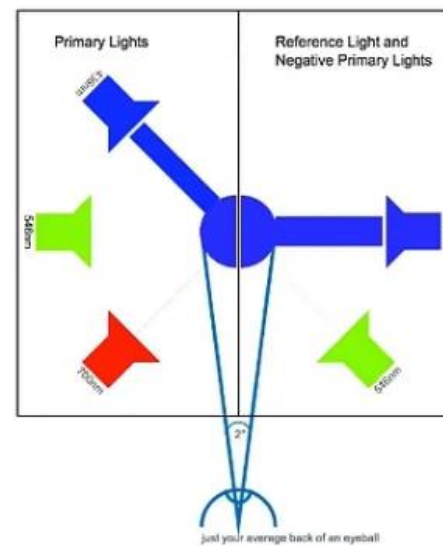


520nm





Great animation at:
<https://medium.com/hipster-color-science/a-beginners-guide-to-colorimetry-401f1830b65a>



**Now that we know color matches to monochromatic spectral lights,
we can compute matches to any color**

Since we know:

$$PM_1 \leftrightarrow e_1 \quad \text{where} \quad M_1 = \begin{bmatrix} m_{1,1} \\ m_{2,1} \\ m_{3,1} \end{bmatrix}$$

using the property of homogeneity: $\alpha PM_1 \leftrightarrow \alpha e_1$

And since also: $PM_2 \leftrightarrow e_2$

Using the property of additivity: $PM_1 + PM_2 \leftrightarrow e_1 + e_2$

Can compute any match

For any color $C = c_1 e_1 + c_2 e_2 + \cdots + c_n e_n$

$$C = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

Because it behaves as a linear system:

$$m_c = \begin{bmatrix} m_{1,c} \\ m_{2,c} \\ m_{3,c} \end{bmatrix} = c_1 \begin{bmatrix} m_{1,1} \\ m_{2,1} \\ m_{3,1} \end{bmatrix} + c_2 \begin{bmatrix} m_{1,2} \\ m_{2,2} \\ m_{3,2} \end{bmatrix} + \cdots = \begin{bmatrix} m_{1,1} & \cdots & m_{1,i} & \cdots & m_{1,n} \\ m_{2,1} & \cdots & m_{2,i} & \cdots & m_{2,n} \\ m_{3,1} & \cdots & m_{3,i} & \cdots & m_{3,n} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = M C$$

$$P m_c \leftrightarrow C \quad \begin{bmatrix} P_{1,1} & P_{2,1} & P_{3,1} \\ P_{1,2} & P_{2,2} & P_{3,2} \\ \vdots & \vdots & \vdots \\ P_{1,n} & P_{2,n} & P_{3,n} \end{bmatrix} \begin{bmatrix} m_{1,1} & \cdots & m_{1,i} & \cdots & m_{1,n} \\ m_{2,1} & \cdots & m_{2,i} & \cdots & m_{2,n} \\ m_{3,1} & \cdots & m_{3,i} & \cdots & m_{3,n} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} \leftrightarrow \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

Why should three primaries let you match any color?

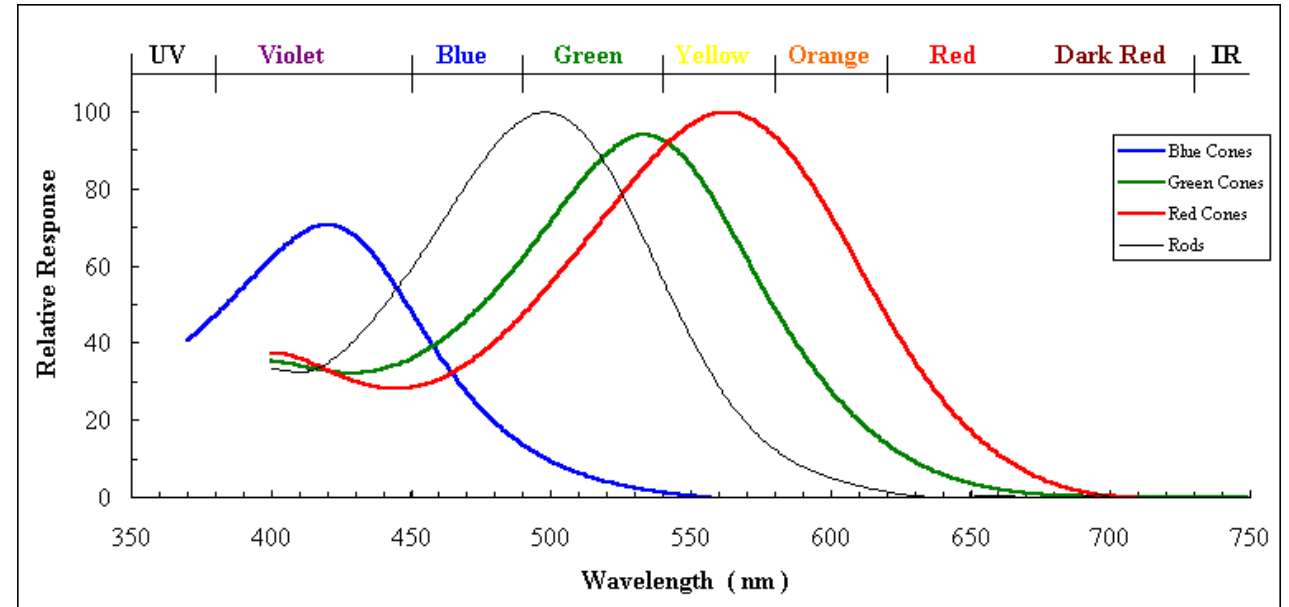
Two basic types of photoreceptors in the retina:

- **rods**: responsible for vision at low light levels, do not mediate color vision
- **cones**: active at higher light levels, capable of color vision

Three basic types of cones in the retina: S-cones, M-cones, L-cones (short, medium, long wavelength)

Different absorption characteristics as a function of wavelength

Many ways to measure these



Cone photocurrent spectral responsivities.

From:
https://www.unm.edu/~toolson/human_cone_response.htm

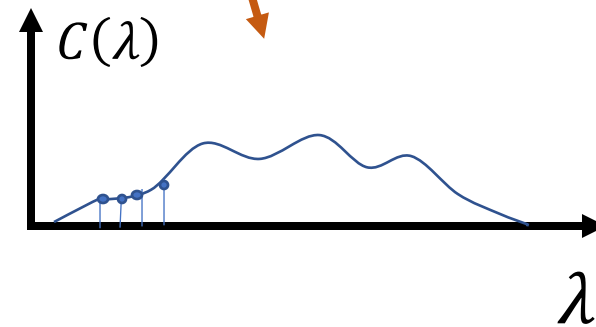
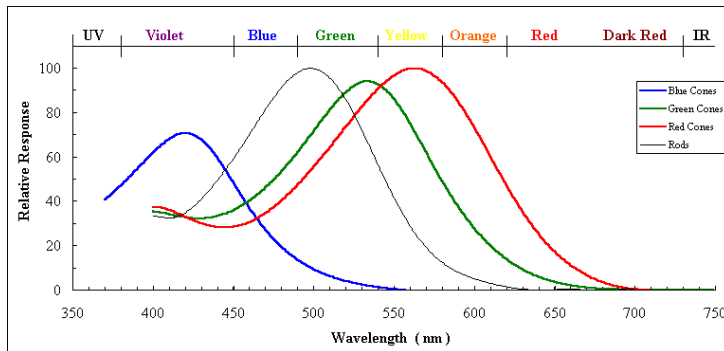
Retinal response

If a patch of retina receives light from a color patch in the viewing field:

$$r = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix} = \underbrace{\begin{bmatrix} R_{1,1} & \dots & R_{1,i} & \dots & R_{1,n} \\ R_{2,1} & \dots & R_{2,i} & \dots & R_{2,n} \\ R_{3,1} & \dots & R_{3,i} & \dots & R_{3,n} \end{bmatrix}}_{\text{Spectral sensitivity of } i^{\text{th}} \text{ cone type}} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = Rc \quad r = Rc$$

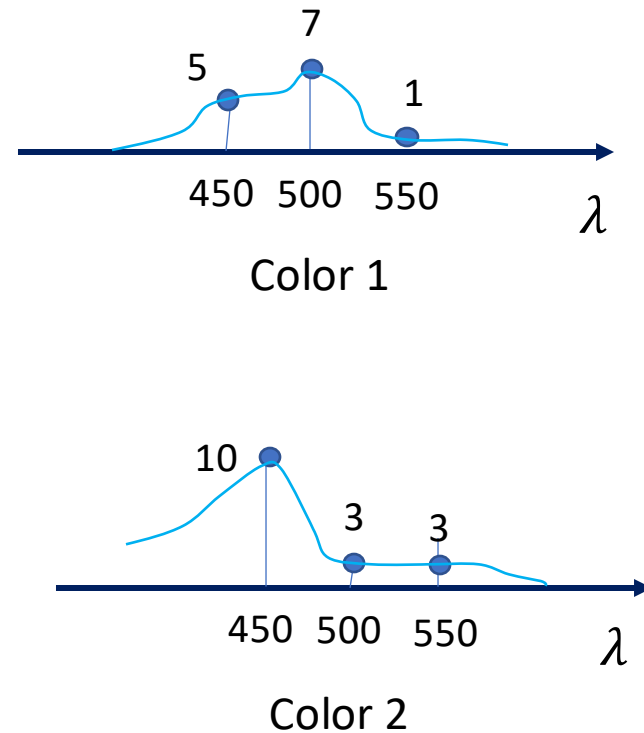
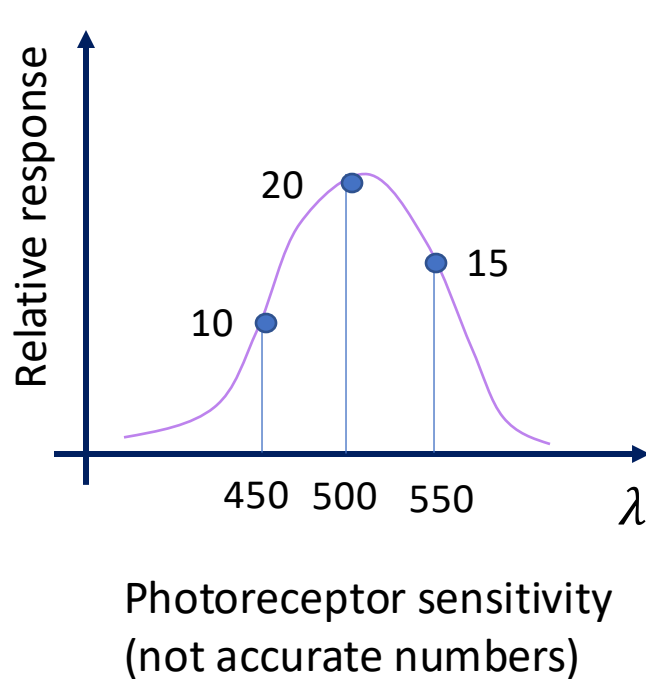
Neural response

Spectral sensitivity of i^{th} cone type



Photoreceptor sensitivity: Toy example 1

In this toy example, there's only one kind of photoreceptor



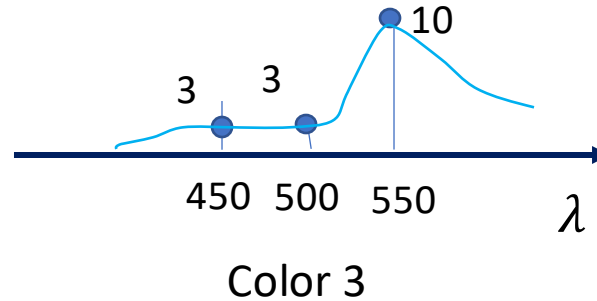
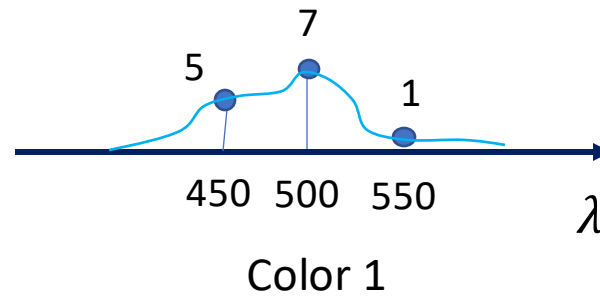
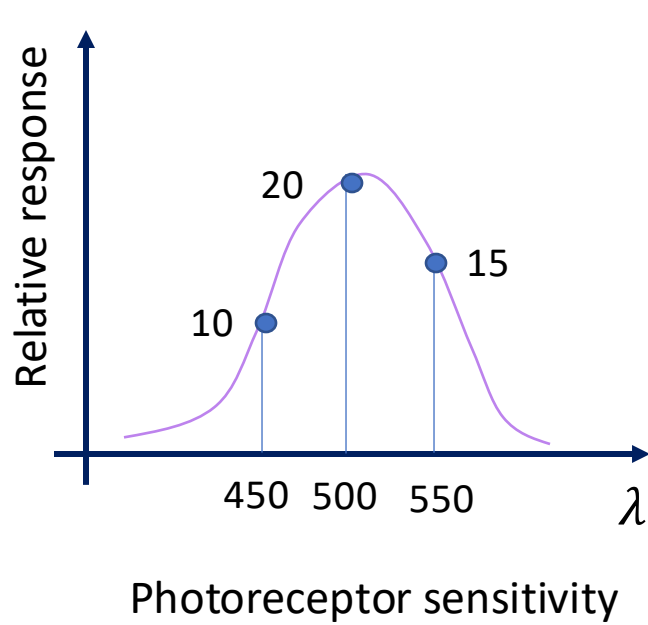
$$\begin{bmatrix} 10 & 20 & 15 \end{bmatrix} \begin{bmatrix} 5 \\ 7 \\ 1 \end{bmatrix} = 205$$

$$\begin{bmatrix} 10 & 20 & 15 \end{bmatrix} \begin{bmatrix} 10 \\ 3 \\ 3 \end{bmatrix} = 205$$

Neural response

In this toy example, the photoreceptor can't tell the difference between Color 1 and Color 2

Photoreceptor sensitivity: Toy example 2



$$\begin{bmatrix} 10 & 20 & 15 \end{bmatrix} \begin{bmatrix} 5 \\ 7 \\ 1 \end{bmatrix} = 205$$

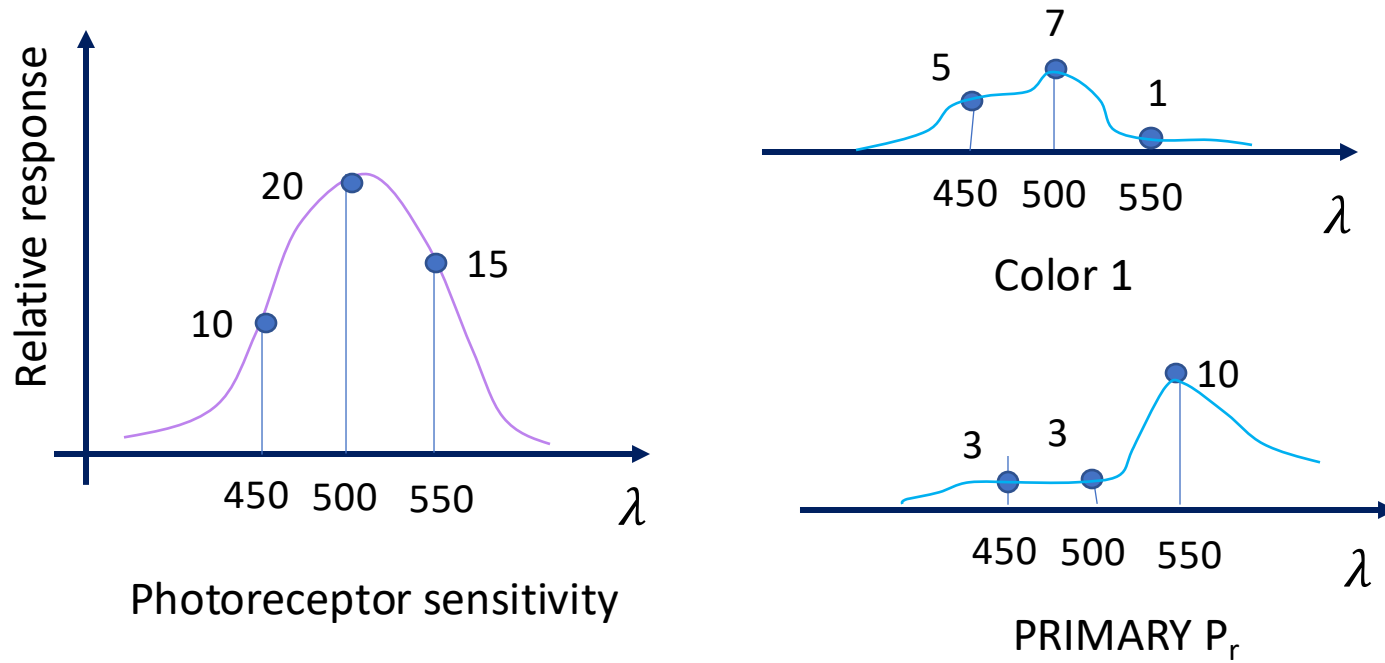
$$\begin{bmatrix} 10 & 20 & 15 \end{bmatrix} \begin{bmatrix} 3 \\ 3 \\ 10 \end{bmatrix} = 240$$

Neural response

Now, the photoreceptor can tell the difference between Color 1 and Color 3

Scotopic viewing conditions (dim light)

In dim light, the 3 cone types do not operate. Only the rods operate, and there is only one kind.



$$\begin{bmatrix} 10 & 20 & 15 \end{bmatrix} \begin{bmatrix} 5 \\ 7 \\ 1 \end{bmatrix} = 205$$

$$\begin{bmatrix} 10 & 20 & 15 \end{bmatrix} \begin{bmatrix} 3 \\ 3 \\ 10 \end{bmatrix} = 240$$

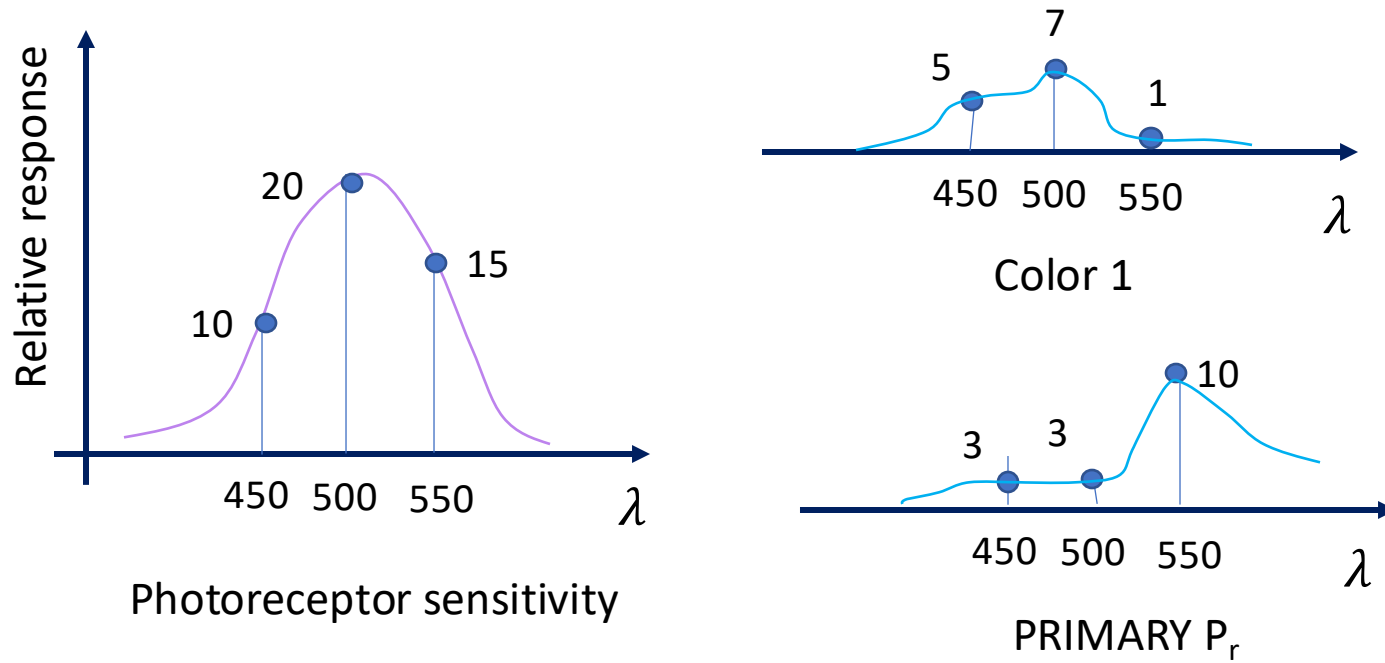
Neural response

Can one adjust P_r to get a match?

Not allowed to change the shape of the distribution, just the overall gain

Scotopic viewing conditions (dim light)

In dim light, the 3 cone types do not operate. Only the rods operate, and there is only one kind.



$$\begin{bmatrix} 10 & 20 & 15 \end{bmatrix} \begin{bmatrix} 5 \\ 7 \\ 1 \end{bmatrix} = 205$$

$$\begin{bmatrix} 10 & 20 & 15 \end{bmatrix} \begin{bmatrix} 3 \\ 3 \\ 10 \end{bmatrix} = 240$$

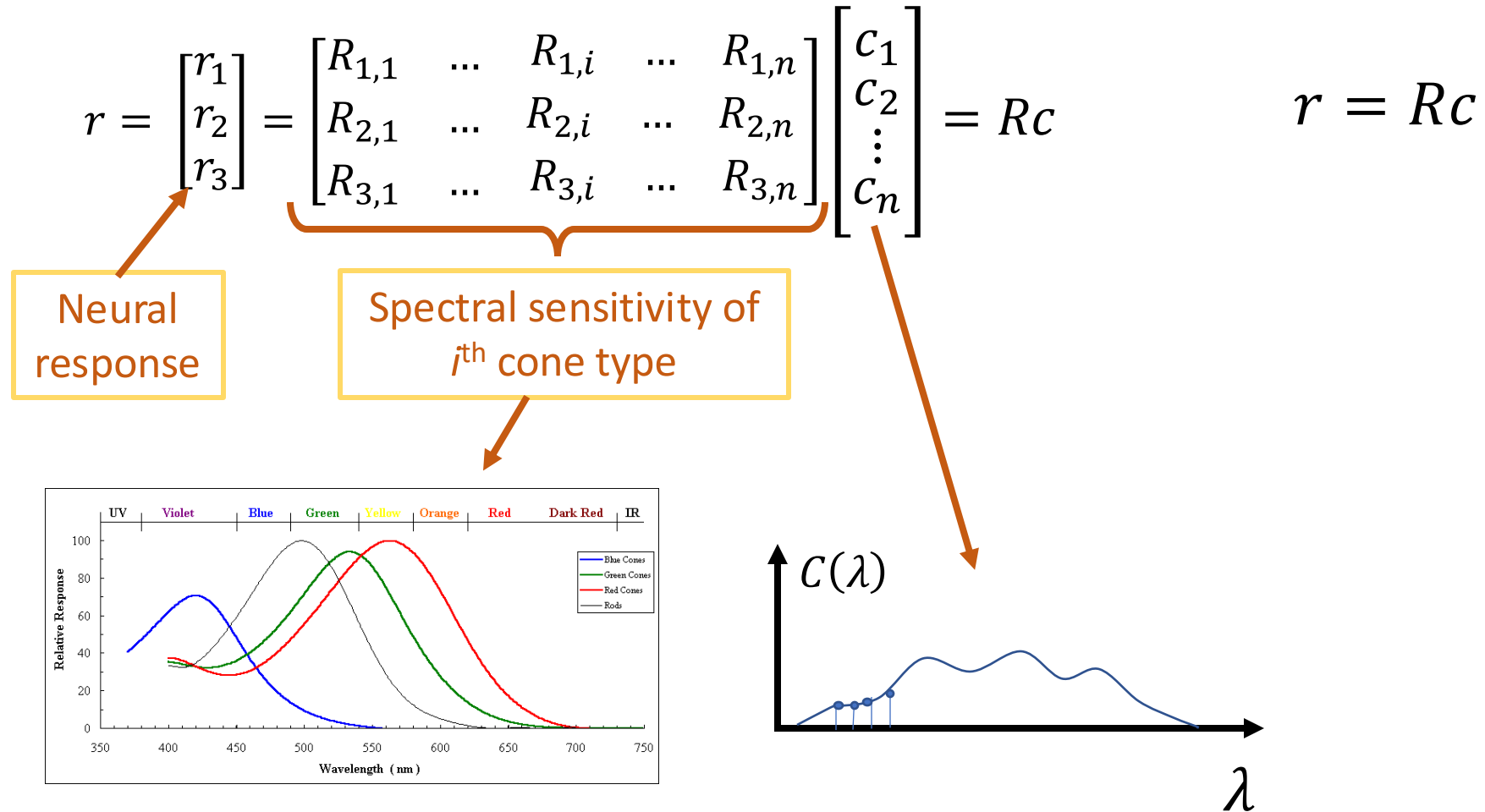
Neural response

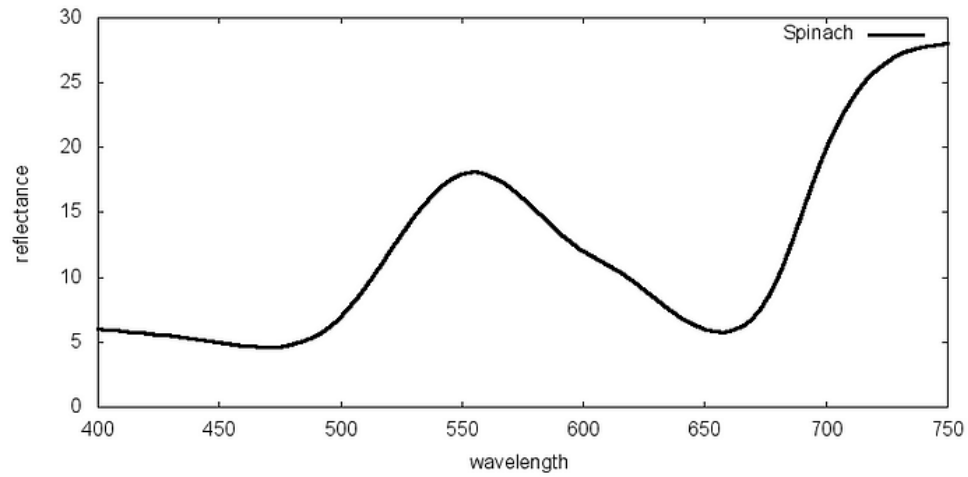
Can one adjust P_r to get a match? Find α so that: $\begin{bmatrix} 10 & 20 & 15 \end{bmatrix} \begin{bmatrix} 3\alpha \\ 3\alpha \\ 10\alpha \end{bmatrix} = 240 \quad \alpha = 205$

In scotopic viewing conditions, any one primary can be used to match any test light

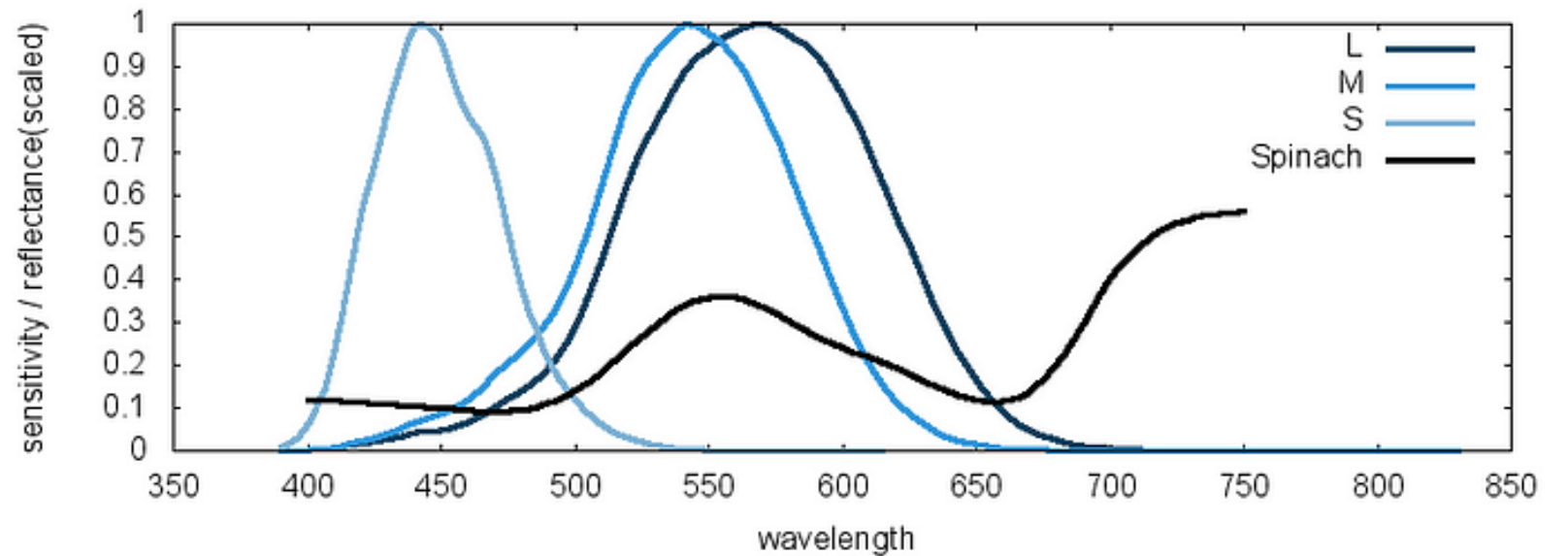
Photopic viewing conditions (bright light)

Three different cone types are operating





The eye's cone system takes a complex spectral distribution and reduces it to three numerical values, representing how much the three cones were stimulated.



Two colors match $\Leftrightarrow$ same neural response

Two colors match:

$$c_1 \leftrightarrow c_2 \quad \Leftrightarrow \quad r = Rc_1 = Rc_2$$

From the color matching experiment, we can find a color match:

$$Pm_c \leftrightarrow c$$

$$r = RPm_c = Rc$$

How do we know it's invertible?

$$m_c = (RP)^{-1}Rc$$

$$\begin{bmatrix} m_{1,c} \\ m_{2,c} \\ m_{3,c} \end{bmatrix} = \begin{bmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix} \begin{bmatrix} R_{1,1} & \dots & R_{1,i} & \dots & R_{1,n} \\ R_{2,1} & \dots & R_{2,i} & \dots & R_{2,n} \\ R_{3,1} & \dots & R_{3,i} & \dots & R_{3,n} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

Apply this to the monochromatic spectral lights:

$$PM_i \leftrightarrow e_i \quad \text{where} \quad M_i = \begin{bmatrix} m_{1,i} \\ m_{2,i} \\ m_{3,i} \end{bmatrix} \quad \text{tristimulus values for matching } e_i$$

$$M_i = (RP)^{-1}Re_i$$

Stack these all up for all of them, $1 \dots n$

$$\begin{bmatrix} m_{1,1} & \dots & m_{1,i} & \dots & m_{1,n} \\ m_{2,1} & \dots & m_{2,i} & \dots & m_{2,n} \\ m_{3,1} & \dots & m_{3,i} & \dots & m_{3,n} \end{bmatrix} = (RP)^{-1}R \begin{bmatrix} 1 & & & & \\ & 1 & & & \\ & & \ddots & & \\ & & & 1 & \\ & & & & 1 \end{bmatrix}$$

$$M = (RP)^{-1}R$$

If any two primaries are a match for the third one, then it's not invertible

Connection between behavior and physiology

matrix of primaries

$$P = \begin{bmatrix} P_{1,1} & P_{2,1} & P_{3,1} \\ P_{1,2} & P_{2,2} & P_{3,2} \\ \vdots & \vdots & \vdots \\ P_{1,n} & P_{2,n} & P_{3,n} \end{bmatrix}$$

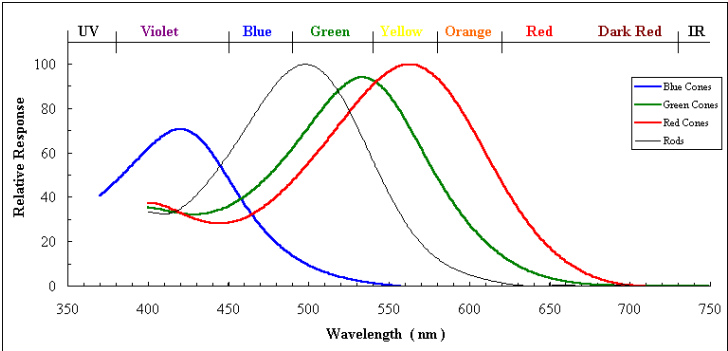
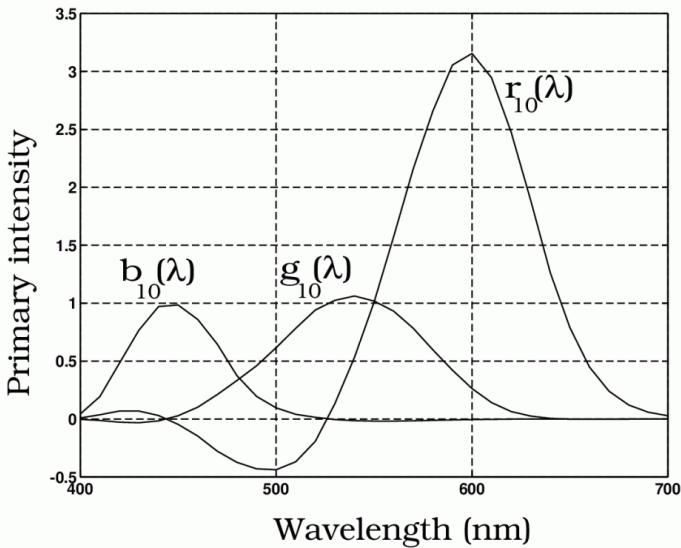
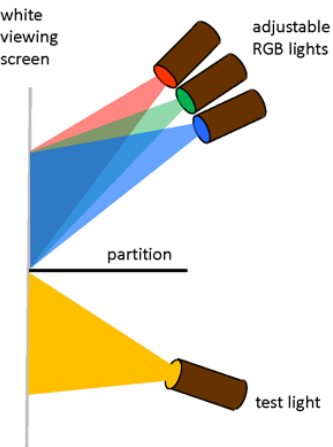
$$M = (RP)^{-1}R$$

$$\begin{bmatrix} m_{1,1} & \dots & m_{1,i} & \dots & m_{1,n} \\ m_{2,1} & \dots & m_{2,i} & \dots & m_{2,n} \\ m_{3,1} & \dots & m_{3,i} & \dots & m_{3,n} \end{bmatrix} = \begin{bmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix} \begin{bmatrix} R_{1,1} & \dots & R_{1,i} & \dots & R_{1,n} \\ R_{2,1} & \dots & R_{2,i} & \dots & R_{2,n} \\ R_{3,1} & \dots & R_{3,i} & \dots & R_{3,n} \end{bmatrix}$$

Color Matching Functions

3x3
matrix

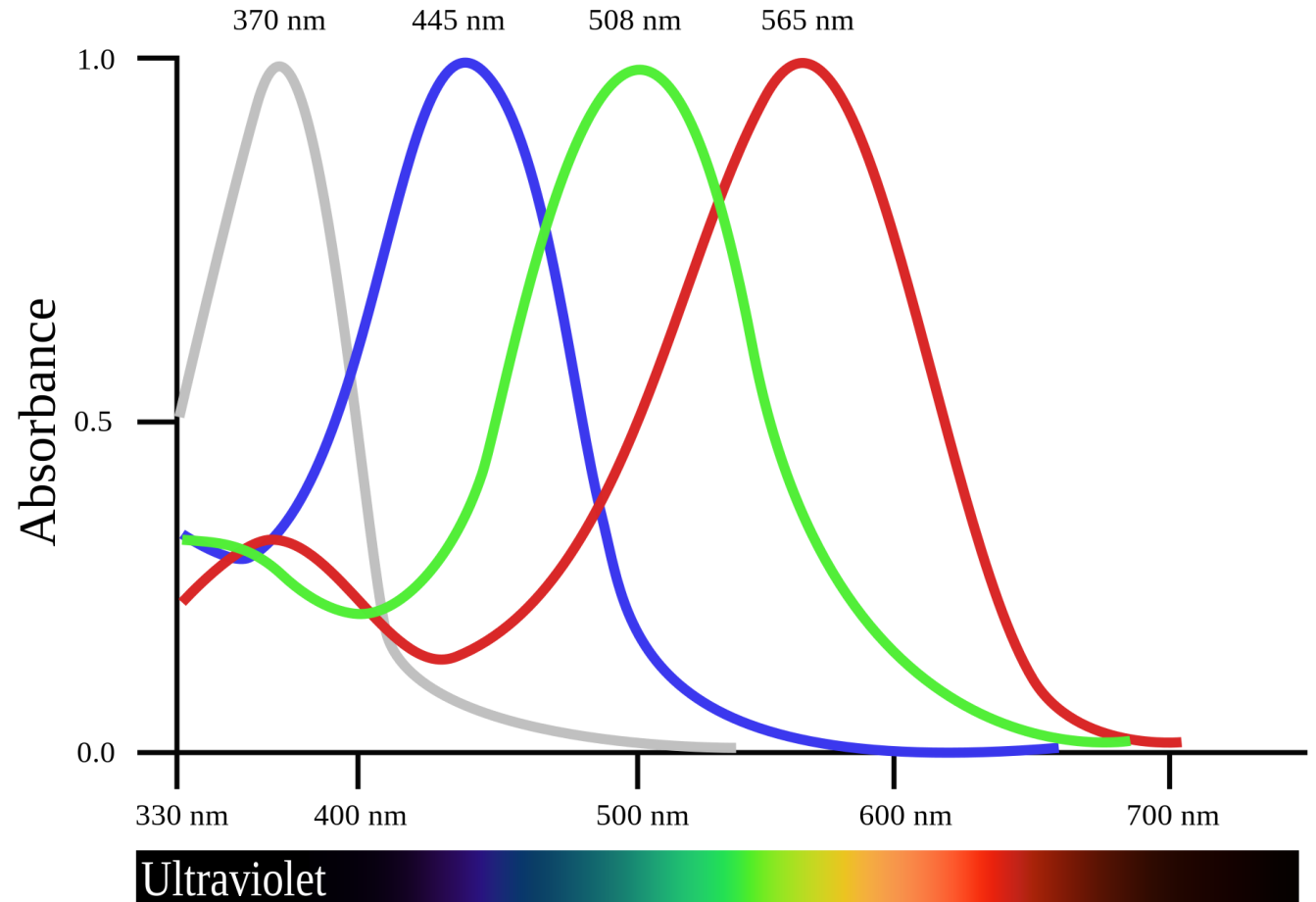
Spectral absorption
functions of the cones



Tetrachromacy

Fun Fact:

- Most birds, fish, and some insects & reptiles are tetrachromatic, with 4 types of cone cell
- In many cases, the 4th photoreceptor allows the animal to perceive UV light



Four pigments in the cone cells of estrildid finches

<https://en.wikipedia.org/wiki/Tetrachromacy>

Matching a color on a TV

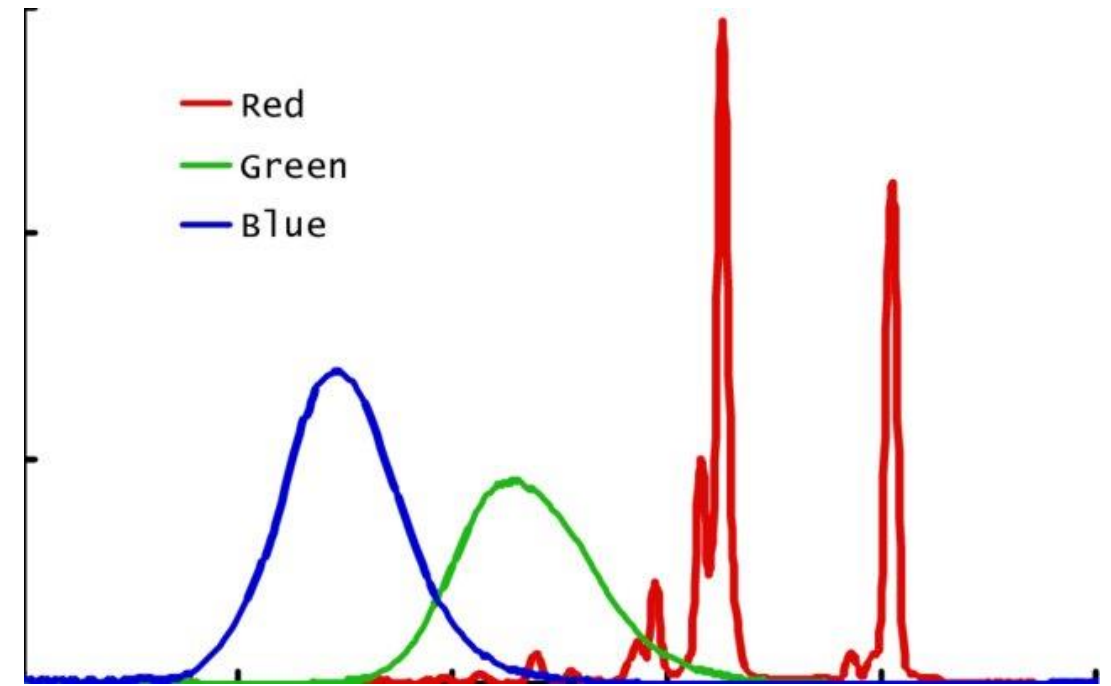
We see some color c in the real world and want to reproduce this on our display

Neural response is: $r = Rc$

Let D represent the spectral power distributions that we have available to display

$$D = \begin{bmatrix} D_{1,1} & D_{2,1} & D_{3,1} \\ D_{1,2} & D_{2,2} & D_{3,2} \\ \vdots & \vdots & \vdots \\ D_{1,n} & D_{2,n} & D_{3,n} \end{bmatrix}$$

Let $g = \begin{bmatrix} g_1 \\ g_2 \\ g_3 \end{bmatrix}$ represent the gains we can apply to them



Typical spectral distributions of P22-series phosphors

Matching a color on a TV

We are trying to attain the neural response $r = Rc$

$$r = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix} = \begin{bmatrix} R_{1,1} & \dots & R_{1,i} & \dots & R_{1,n} \\ R_{2,1} & \dots & R_{2,i} & \dots & R_{2,n} \\ R_{3,1} & \dots & R_{3,i} & \dots & R_{3,n} \end{bmatrix} \begin{bmatrix} D_{1,1} & D_{2,1} & D_{3,1} \\ D_{1,2} & D_{2,2} & D_{3,2} \\ \vdots & \vdots & \vdots \\ D_{1,n} & D_{2,n} & D_{3,n} \end{bmatrix} \begin{bmatrix} g_1 \\ g_2 \\ g_3 \end{bmatrix} = RDg$$

Choose g to match that light

$$RDg = Rc$$

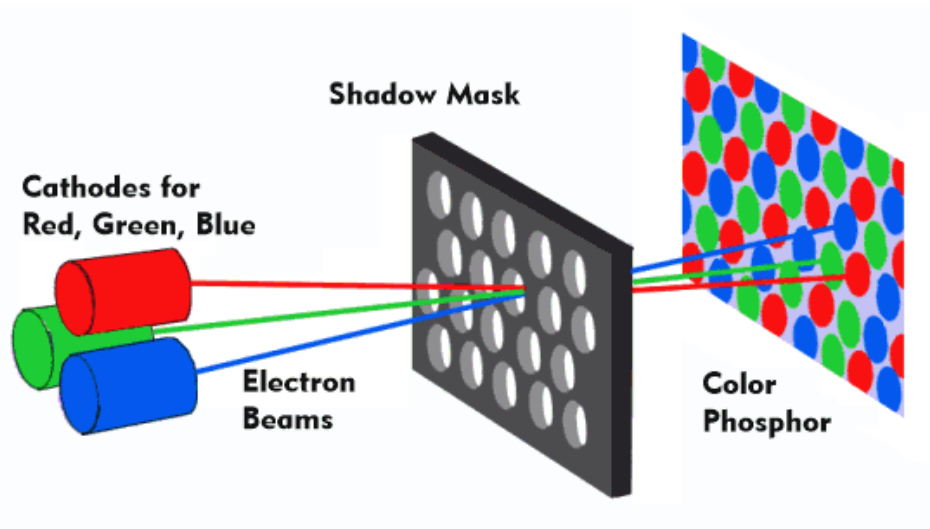
$$g = (RD)^{-1}Rc$$

$0 \leq g \leq Max$ constrains the colors that you can match

No unplugging the primary to move it
to the other side of the box

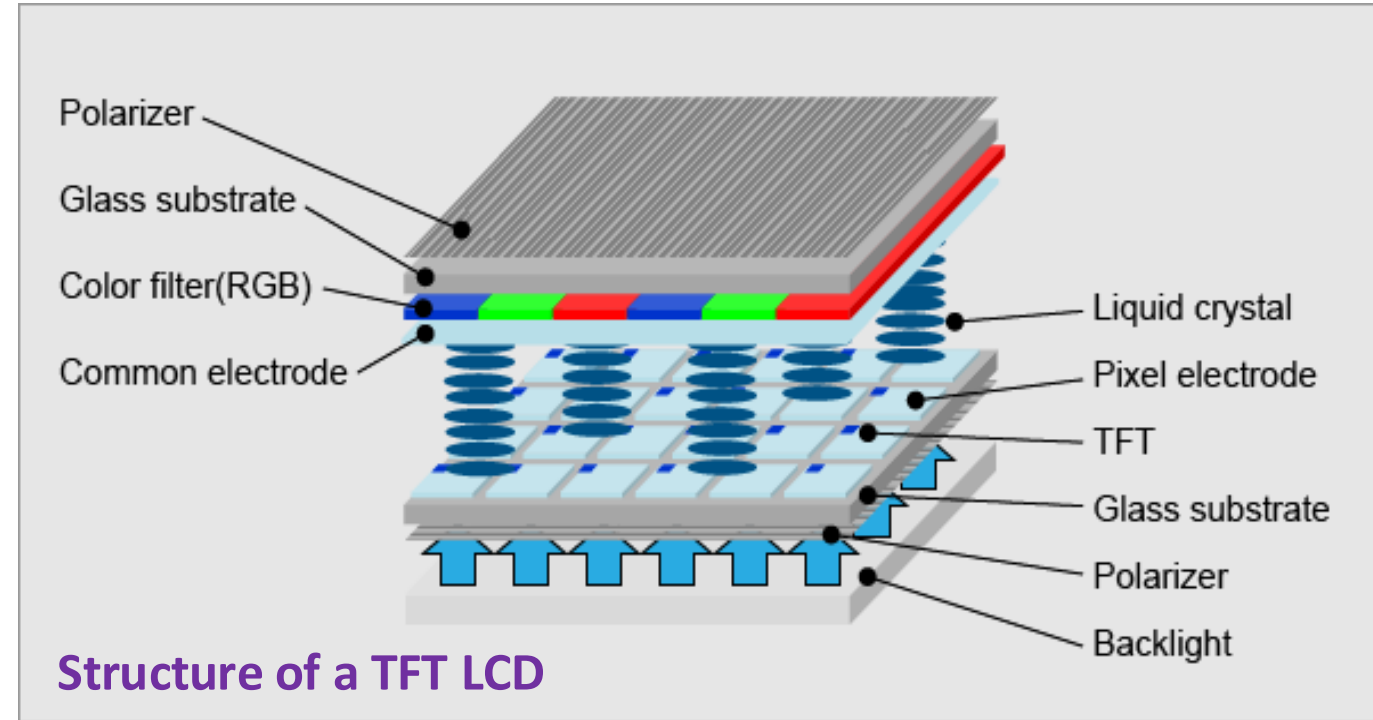
CRT and LCD displays:

RGB additive colors



- Color phosphors have different spectra
- Electron guns shoot electrons at the phosphors, to make them phosphoresce
- Shadow mask controls which phosphor dots get hit

From: <https://www.j-display.com/english/technology/lcdbasic.html>



- TFT = Thin Film Transistor
- Controls the voltage applied to the liquid crystals
- A change in voltage changes the transmittance of the panel
- Changes the quantity of light passing from the backlight to the front surface of the display

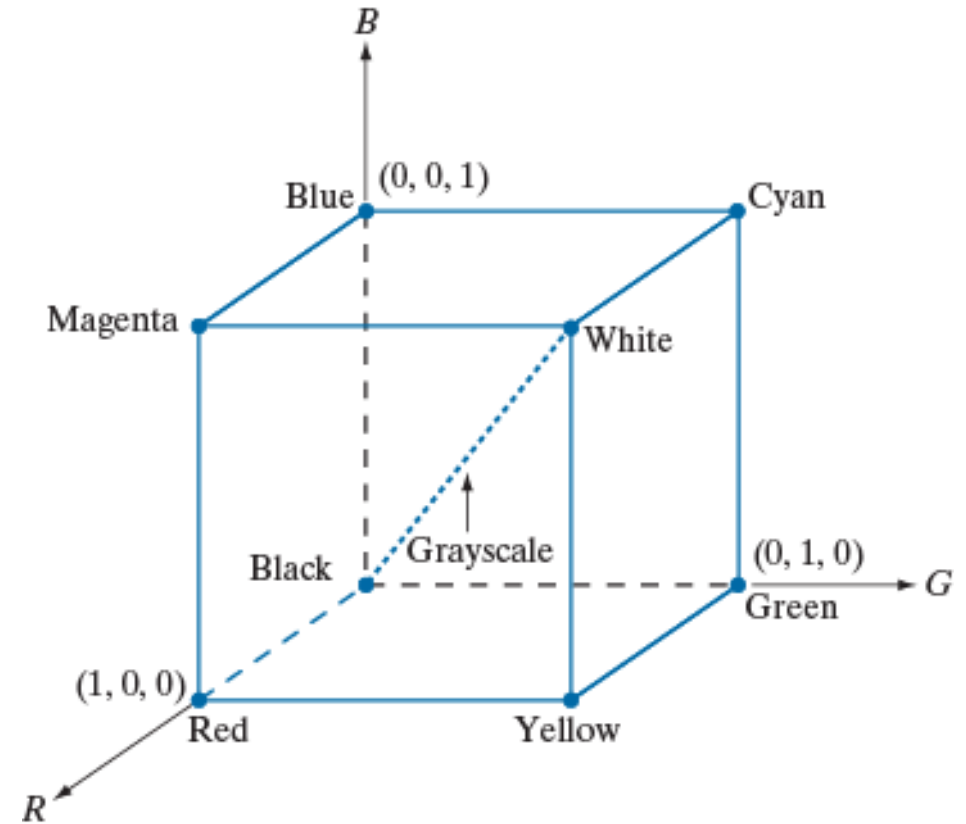
Color Cube

One way to think about displayable colors is with a color cube

Call the three primaries Red, Green, Blue

Black = (0,0,0), White=(1,1,1)

No negative tristimulus values



Trichromatic Coefficients

- Suppose color stimulus C has tristimulus values R,G,B
- Color stimulus 2C (with twice the intensity) has tristimulus values 2R, 2G, 2B

Define:

$$r = \frac{R}{R + G + B} \quad g = \frac{G}{R + G + B} \quad b = \frac{B}{R + G + B}$$

These are called *trichromatic coefficients*

How do trichromatic coefficients of color signal 2C relate to those for C?

Notice: $r + g + b = 1$

When the intensity of a color is not of interest, then only 2 coordinates are needed to specify a match

Chromaticity coordinates: r, g



Any two of them could have been chosen,
3rd one is determined by $r + g + b = 1$

Note: Previous slides used:

- r to represent the neural response (3-tuple)
- g to represent the gains for the monitor display (3-tuple)

Now, r, g are scalars and are not the same things as before!

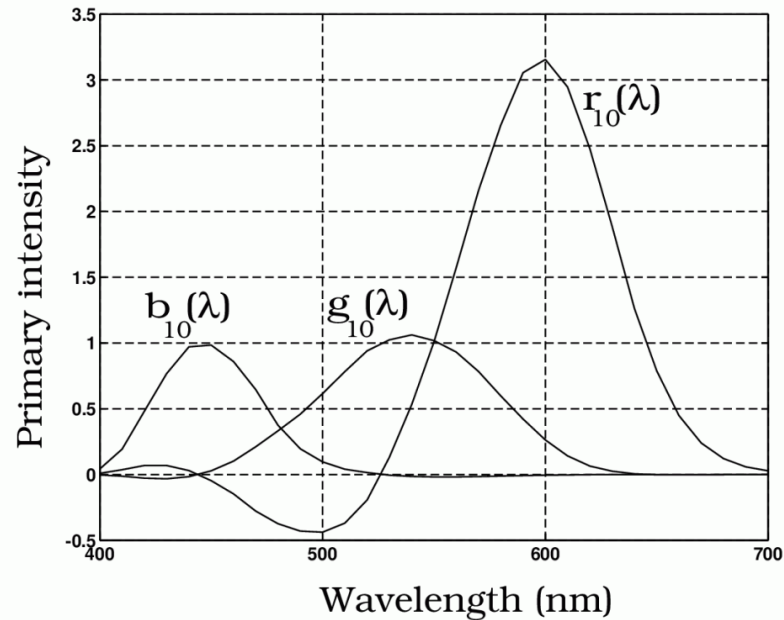
Chromaticity Diagram from coordinates r,g

$P_1 P_2 P_3$
Match e_i

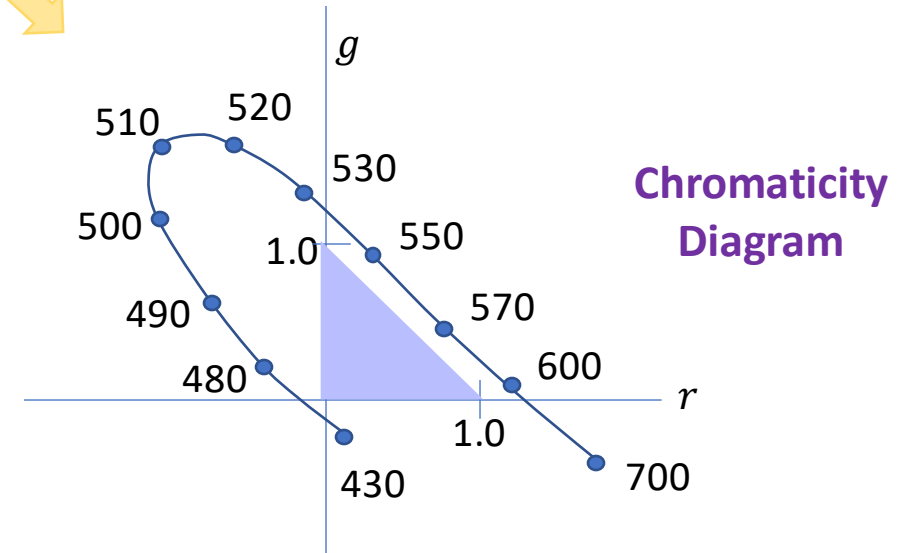
$R_i G_i B_i$
Tristimulus
Values

$r_i g_i b_i$
Trichromatic
coefficients

$$r_i = \frac{R_i}{R_i + G_i + B_i} \quad g_i = \frac{G_i}{R_i + G_i + B_i}$$



Color
Matching
Functions



Chromaticity coordinates of mixtures

Suppose color C_1 has tristimulus values X_1, Y_1, Z_1

Color C_2 has tristimulus values X_2, Y_2, Z_2

We form an additive mixture $C = C_1 + C_2$

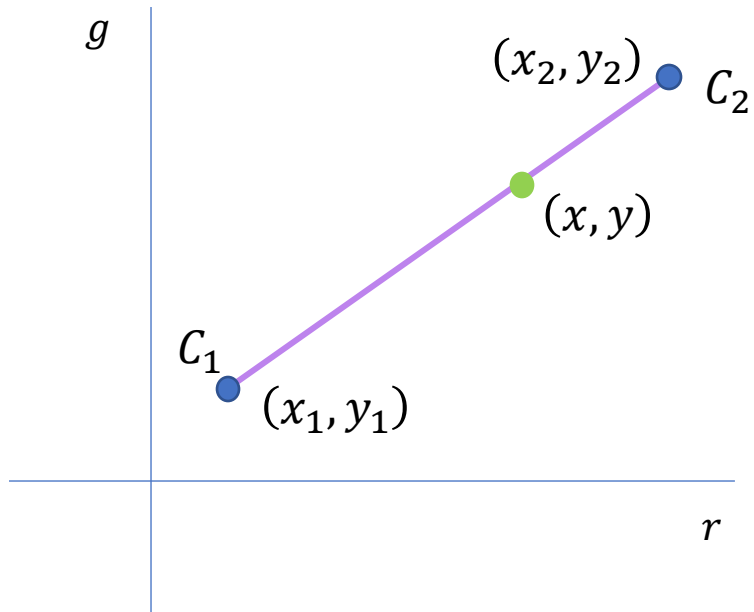
It can be shown that the point (x, y) lies on the line segment that connects (x_1, y_1) and (x_2, y_2)

Color	C_1	C_2	$C = C_1 + C_2$
Tristimulus Values	X_1, Y_1, Z_1	X_2, Y_2, Z_2	$X_1 + X_2, Y_1 + Y_2, Z_1 + Z_2$
Chromaticity Coordinates	$x_1 = \frac{X_1}{X_1 + Y_1 + Z_1},$ $y_1 = \frac{Y_1}{X_1 + Y_1 + Z_1}$	$x_2 = \frac{X_2}{X_2 + Y_2 + Z_2},$ $y_2 = \frac{Y_2}{X_2 + Y_2 + Z_2}$	$x = \frac{X_1 + X_2}{X_1 + X_2 + Y_1 + Y_2 + Z_1 + Z_2},$ $y = \frac{Y_1 + Y_2}{X_1 + X_2 + Y_1 + Y_2 + Z_1 + Z_2}$

Chromaticity coordinates of mixtures

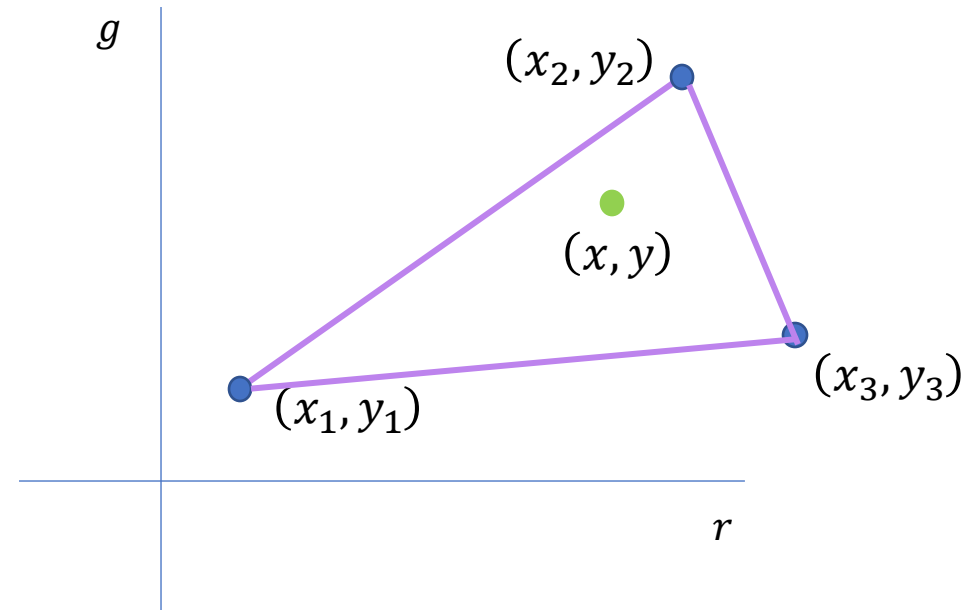
$$C = \alpha C_1 + \beta C_2$$

$$\alpha \geq 0, \beta \geq 0$$



This generalizes to mixtures of 3 colors:

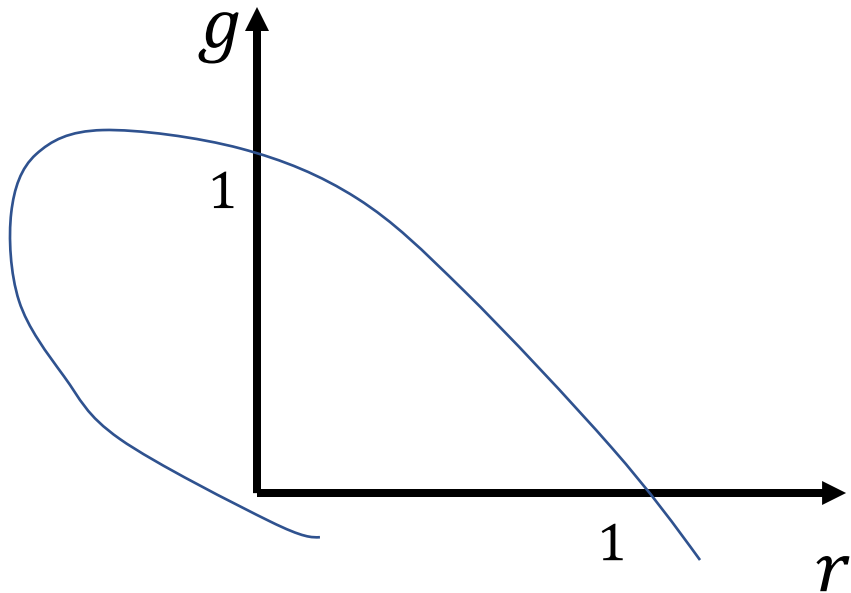
$$C = \alpha C_1 + \beta C_2 + \gamma C_3$$



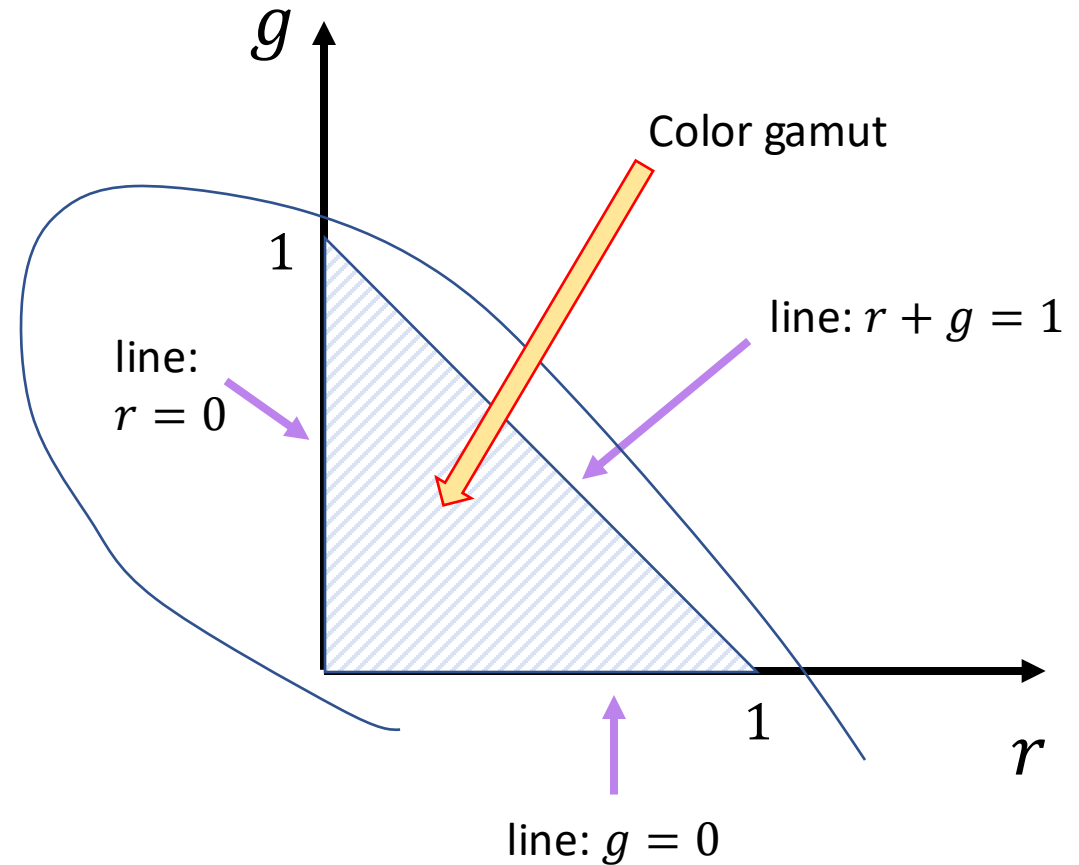
Color C lies somewhere in the triangle

Color Gamut

1. Take some primaries P_1, P_2, P_3
2. Do color matching experiment, plot chromaticity results

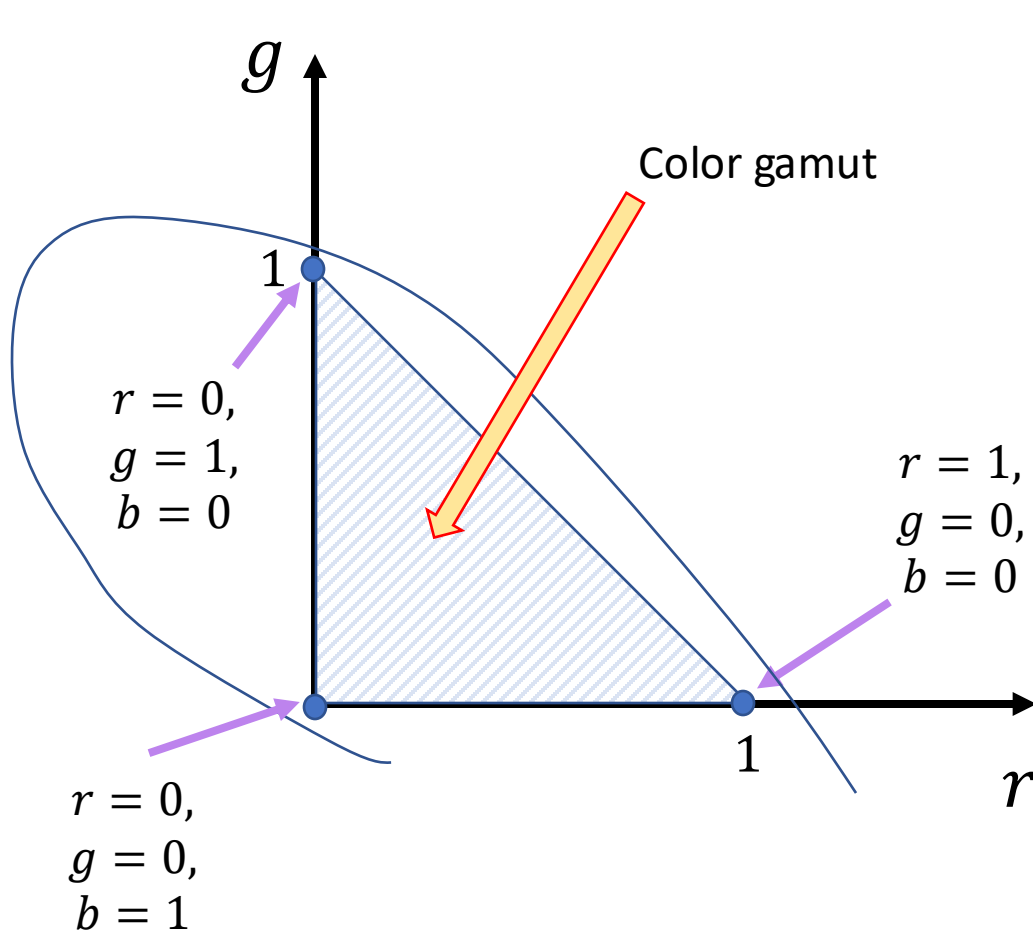


Then: This triangle is a representation of all the colors which can be displayed with additive combinations of these primaries



Color Gamut

Claim 1: Colors inside the triangle can be displayed



$$r = \frac{R}{R + G + B} \quad g = \frac{G}{R + G + B} \quad b = \frac{B}{R + G + B}$$

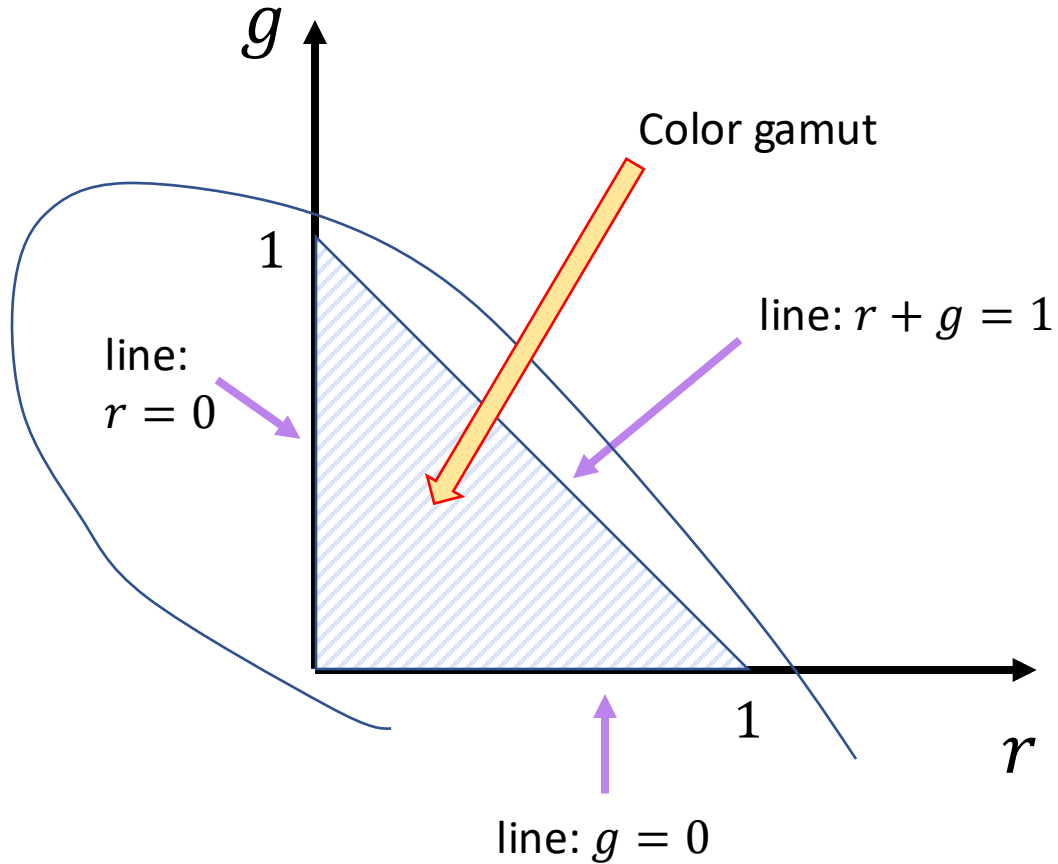
The corners correspond to displayable colors:

- $r = 1, g = 0, b = 0 \Rightarrow \begin{cases} R = \text{any positive value} \\ G = B = 0 \end{cases}$

The corners correspond to the three primaries. Points inside the triangle are additive combinations of them.

Color Gamut

Claim 2: No colors outside the triangle can be displayed



- $r < 0 \Rightarrow \begin{cases} R < 0, & \text{or} \\ R + B + G < 0 \end{cases}$

At least one of R, G, B is less than zero,
 $\Rightarrow$ not displayable

- $g < 0 \Rightarrow \begin{cases} G < 0, & \text{or} \\ R + B + G < 0 \end{cases}$

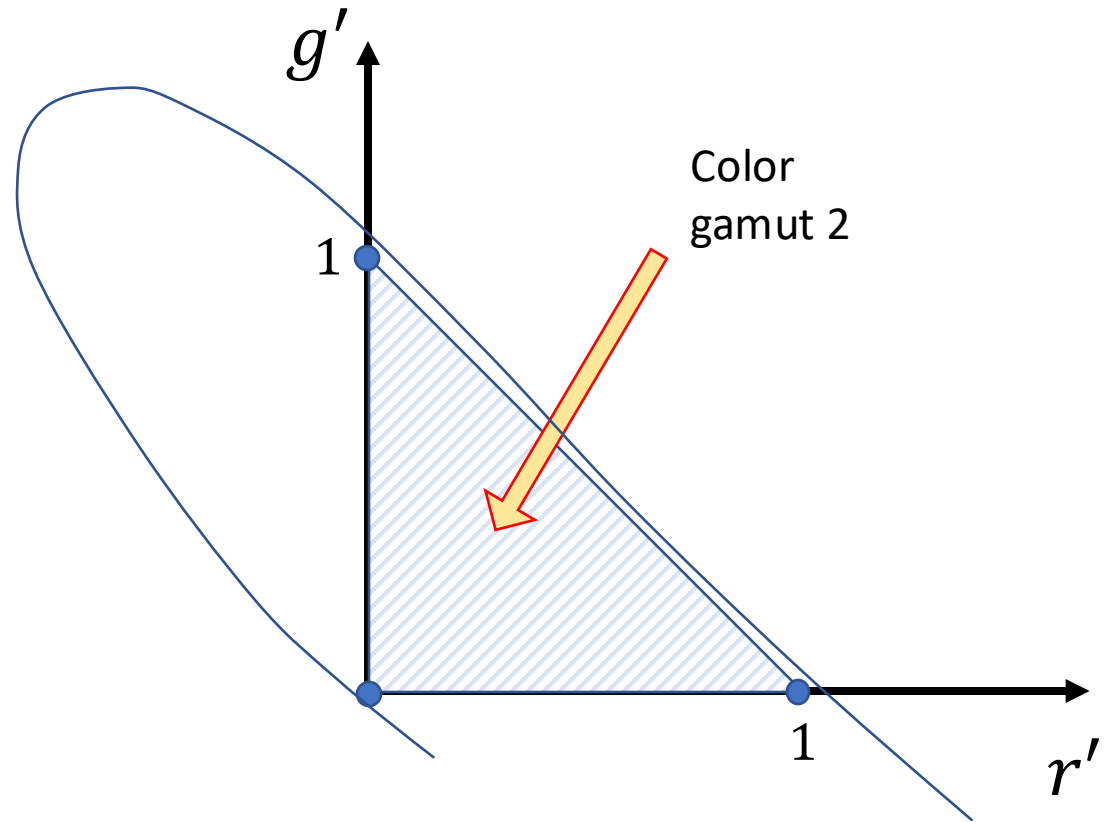
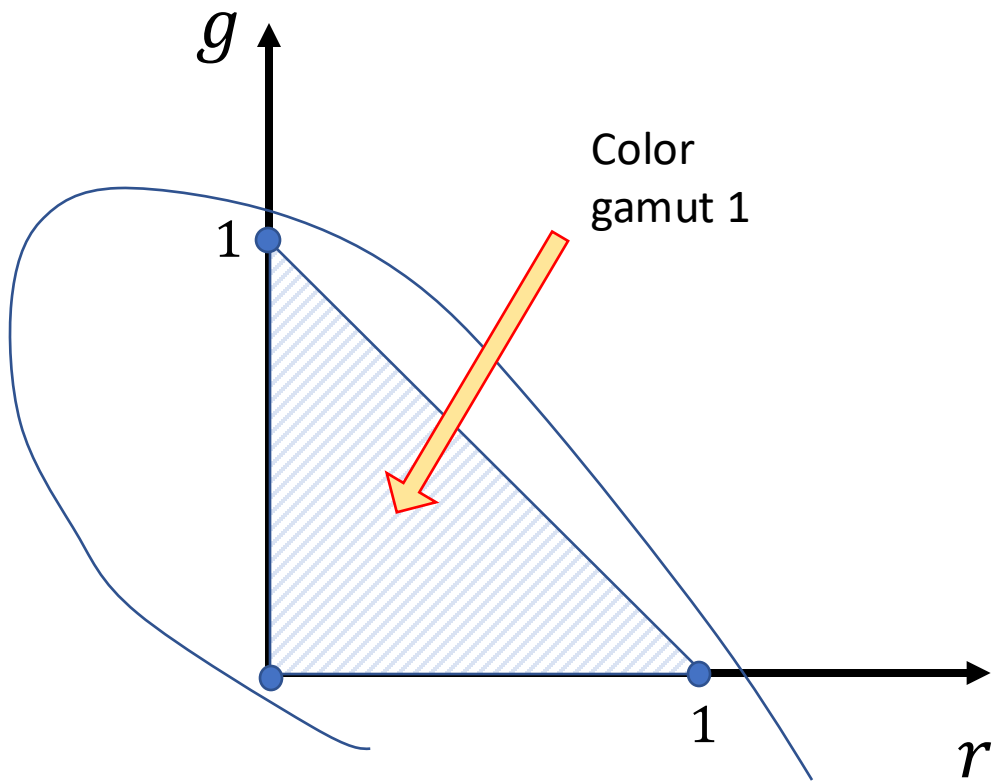
Same thing.

- $r + g > 1 \Rightarrow b < 0 \Rightarrow \begin{cases} B < 0, & \text{or} \\ R + B + G < 0 \end{cases}$

Color Gamuts from different sets of Primaries

- Using different sets of primaries: Which one is better?
- It's not the same coordinate system!

How do we get them into the same coordinate system?



Change of Primaries:

Given primaries P , get color matching functions M

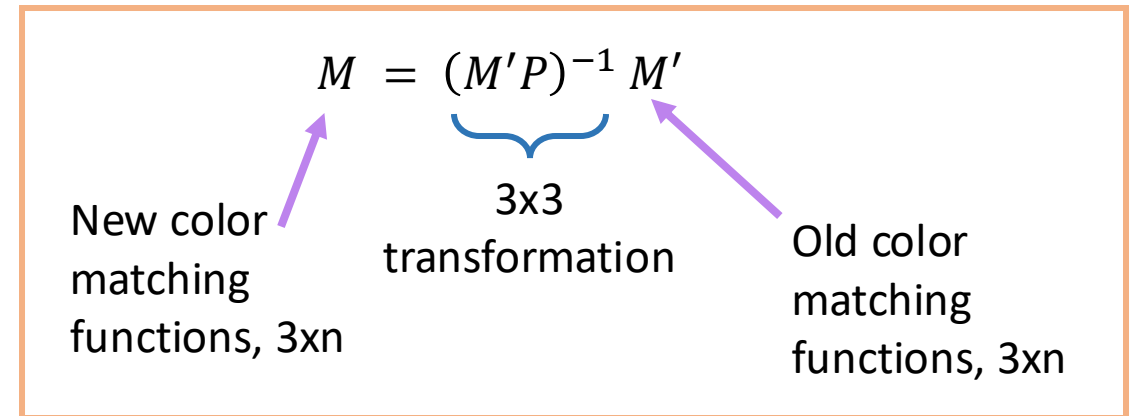
$$M = (RP)^{-1}R$$

Given primaries P' , get color matching functions M'

$$M' = (RP')^{-1}R$$

$$M' = (RP')^{-1}R \Rightarrow R = (RP')M'$$

$$\begin{aligned} M &= (RP)^{-1}R \\ &= ((RP')M'P)^{-1}(RP')M' \\ &= (M'P)^{-1}(RP')^{-1}(RP')M' \\ &= (M'P)^{-1}M' \end{aligned}$$



For a color C , let:

- m_c denote its tristimulus values relative to primaries P
 - m'_c denote its tristimulus values relative to primaries P'
- $\longrightarrow m_c = (M'P)^{-1} m'_c$

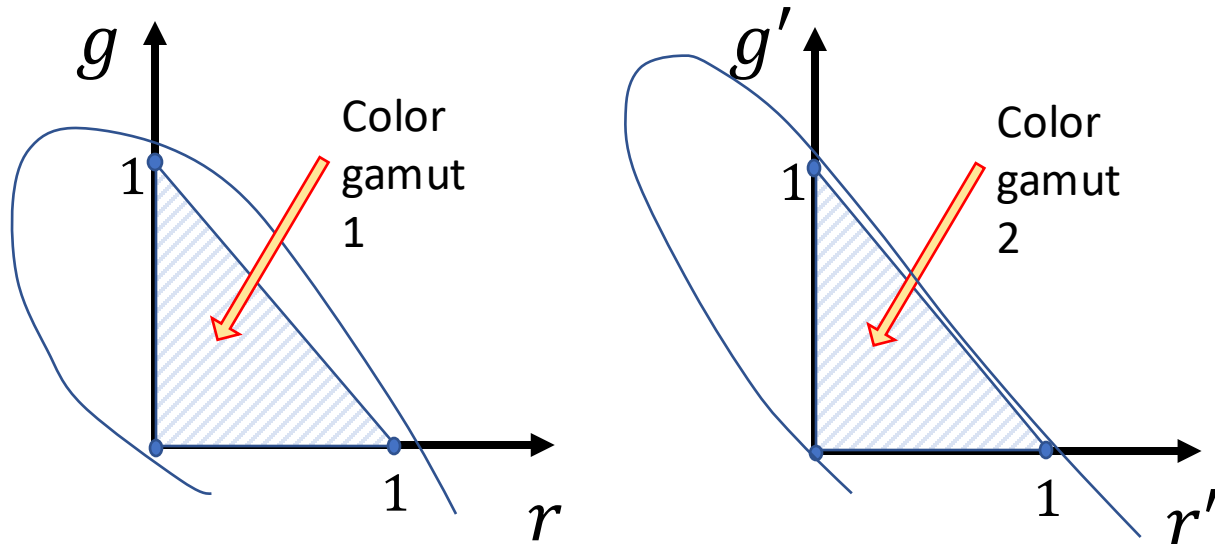
Converting primaries

If two companies create different phosphors for monitor display, color coordinates can be easily converted

If two researchers study color matching on different populations with different primaries, color coordinates can be easily converted

$$\begin{bmatrix} R_N \\ G_N \\ B_N \end{bmatrix} = \begin{bmatrix} 0.842 & 0.156 & 0.091 \\ -0.129 & 1.320 & -0.203 \\ 0.008 & -0.069 & 0.897 \end{bmatrix} \begin{bmatrix} R_c \\ G_c \\ B_c \end{bmatrix}$$

Now we can return to comparing color gamuts from different sets of primaries



- Which set is better?
- We can convert them to another coordinate system

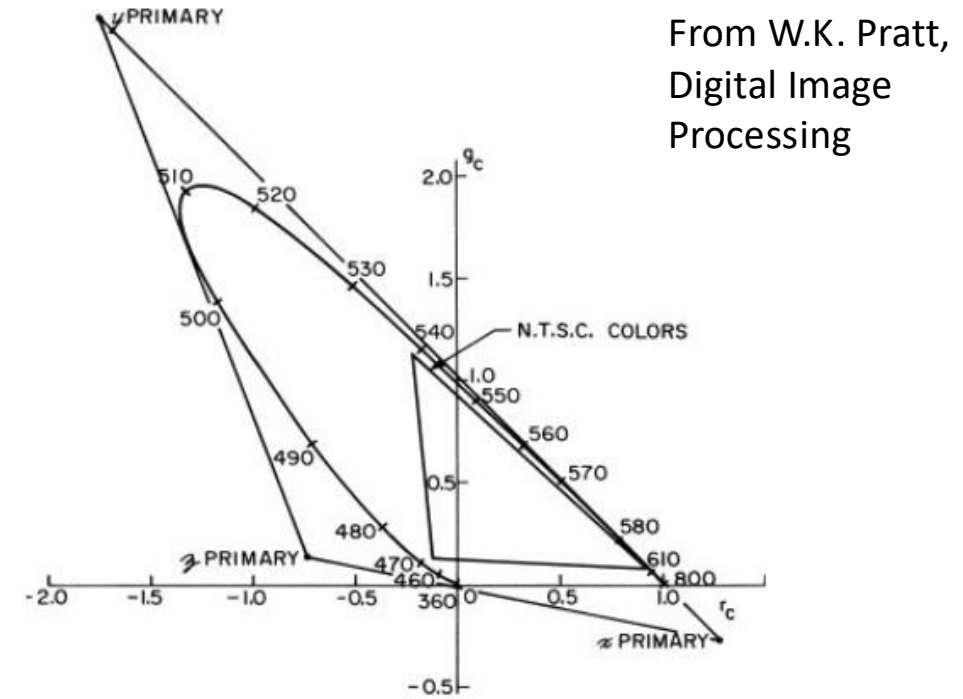


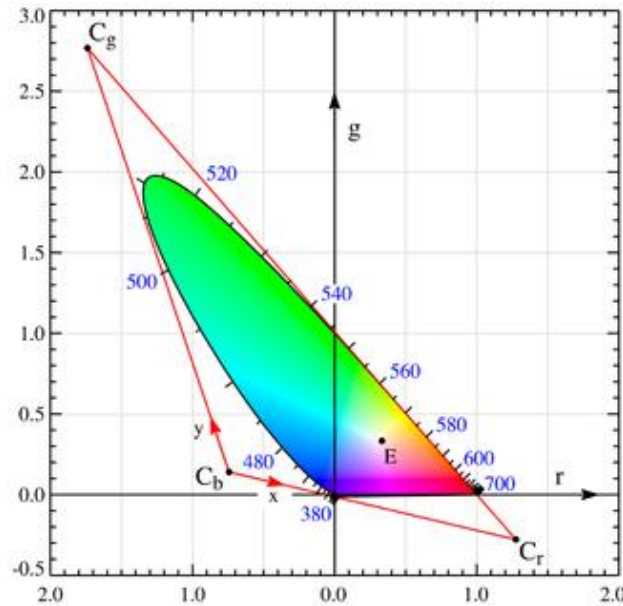
FIGURE 3.5-3. Chromaticity diagram for CIE spectral primary system.

Idea: Choose a coordinate system that makes comparisons convenient

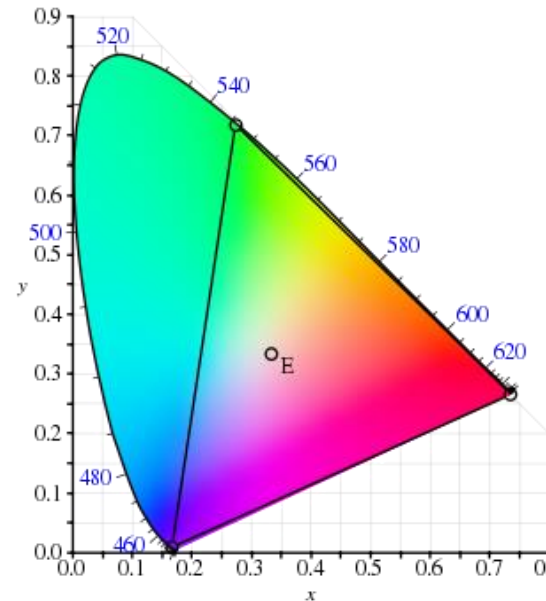
Everything in the positive quadrant

CIE XYZ Color Coordinate System

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} 0.49000 & 0.32000 & 0.2 \\ 0.17697 & 0.81240 & 0.01063 \\ 0 & 0.01 & 0.99000 \end{bmatrix} \begin{bmatrix} R_c \\ G_c \\ B_c \end{bmatrix}$$



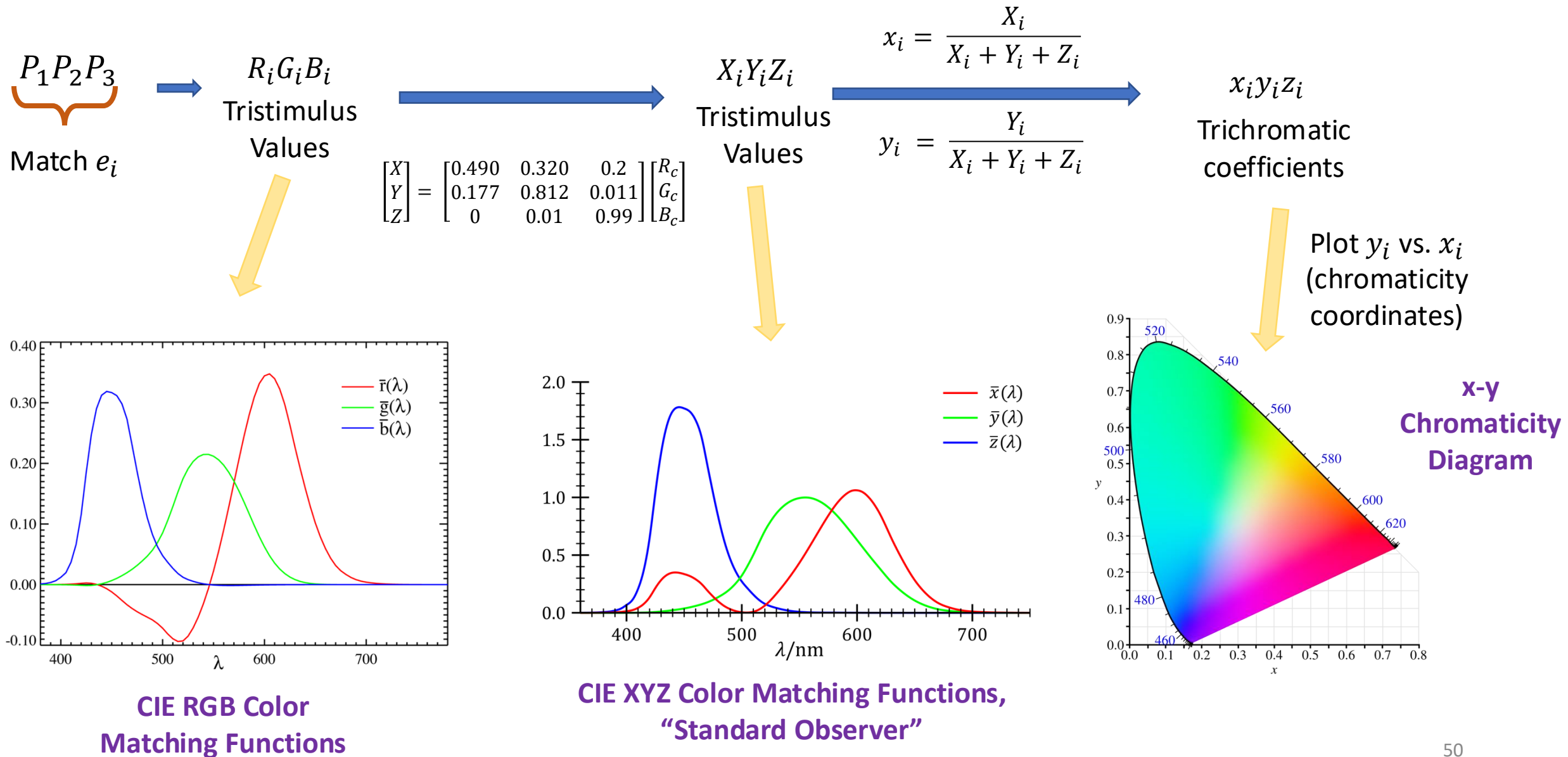
r-g coordinates



x-y coordinates

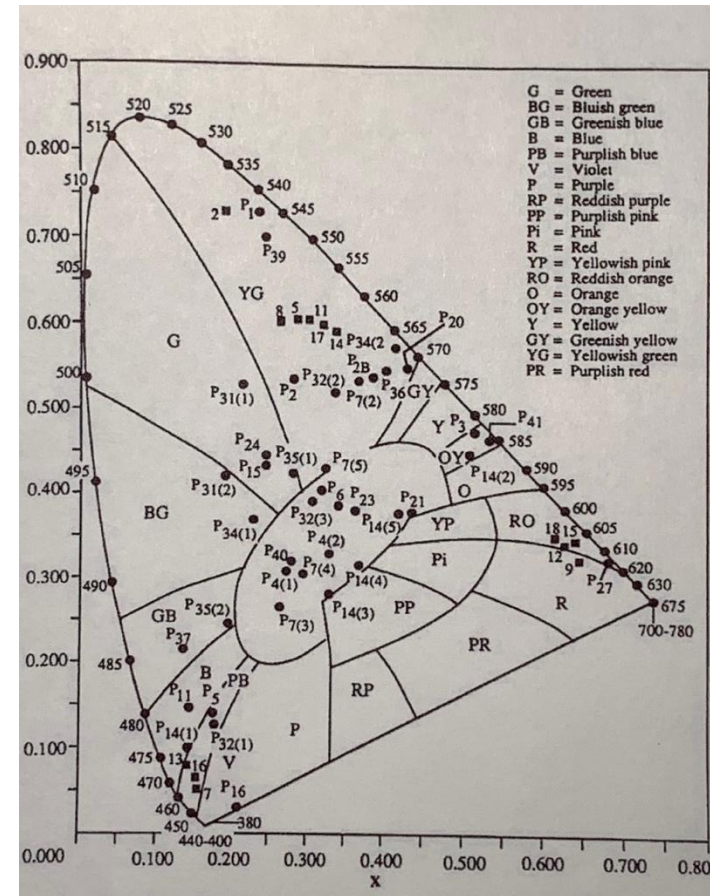
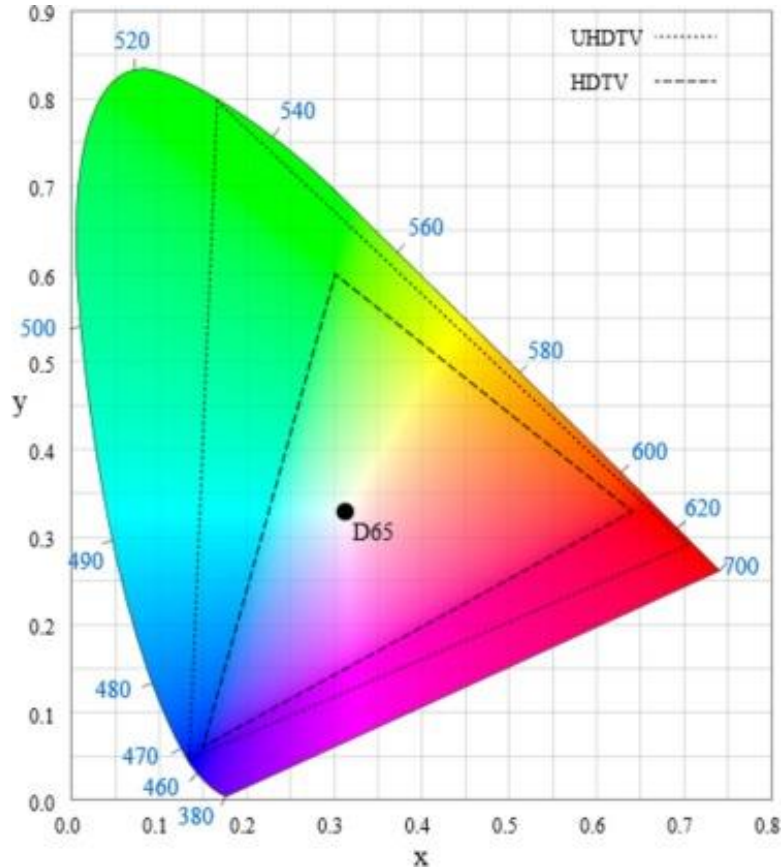
- Mathematical exercise that rotates the chromaticity plot so that everything is in the positive quadrant
- X, Y, Z are always positive
- x, y, z are always positive
- Known as the CIE XYZ system
- Developed in 1931, still the main coordinate system used today for referencing other color systems
- The primaries that correspond to this system are physically unrealizable

x-y Chromaticity Diagram



Example comparison of two RGB color gamuts

Many different CRT phosphors (different spectra) one could use for display



Many factors in choosing CRT phosphors for display:

- Color gamut
- Cost
- Decay time
- Efficiency

From: David R. Bull, Fan Zhang, in [Intelligent Image and Video Compression \(Second Edition\)](#), 2021

Many different RGB systems

1. CIE RGB: (R_C, G_C, B_C) The monochromatic spectral primaries, used in the 1920's for the original color matching experiments
 - Used Red=700nm, Green=546.1nm, Blue=435.8nm
2. NTSC RGB: (R_N, G_N, B_N) National Television Systems Committee receiver primary system
 - Based on CRTs from the 1950s

$$\begin{bmatrix} R_N \\ G_N \\ B_N \end{bmatrix} = \begin{bmatrix} 0.842 & 0.156 & 0.091 \\ -0.129 & 1.320 & -0.203 \\ 0.008 & -0.069 & 0.897 \end{bmatrix} \begin{bmatrix} R_C \\ G_C \\ B_C \end{bmatrix}$$

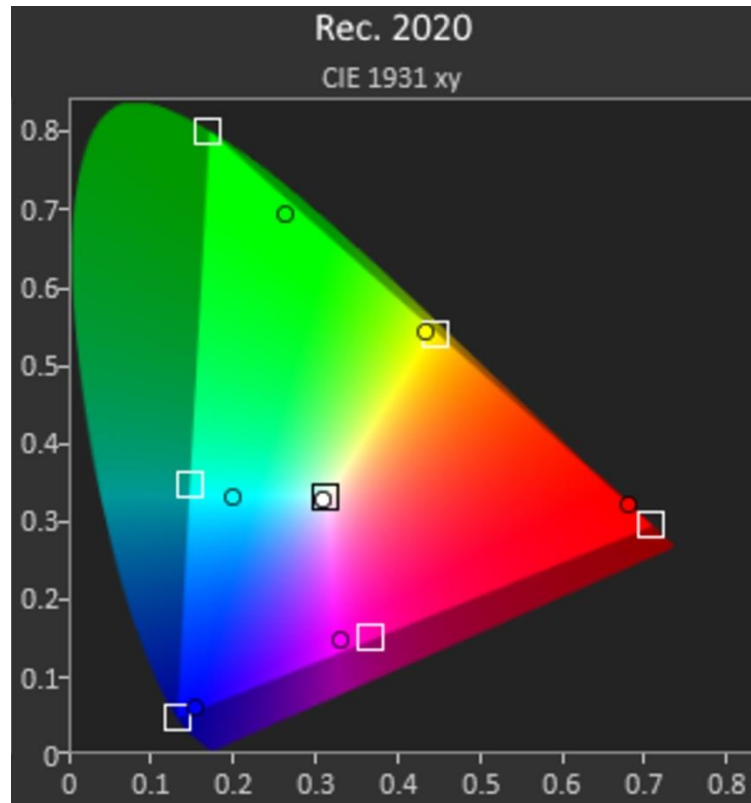
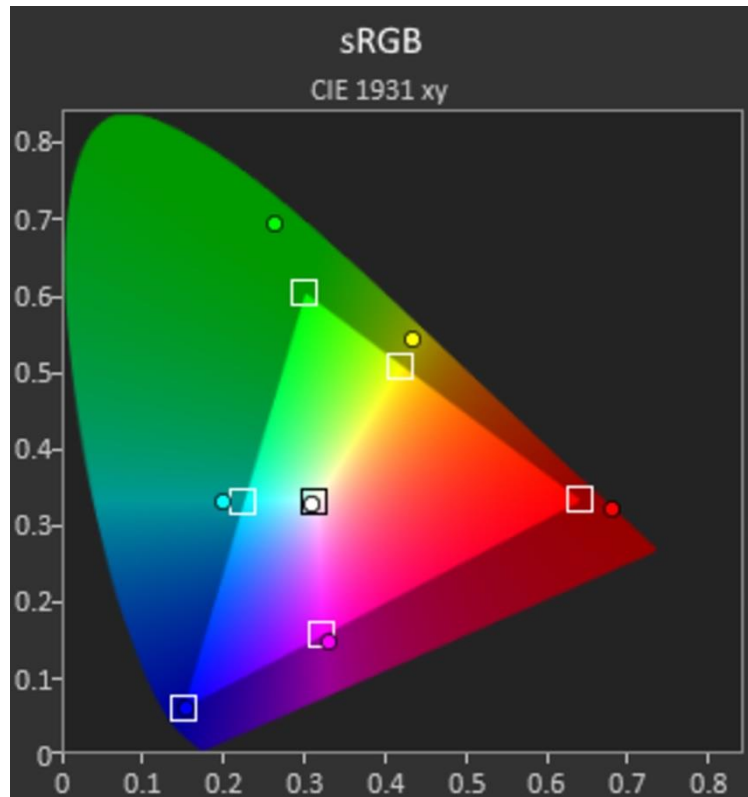
3. sRGB: Intended for Internet use
 - Based on CRTs from the 1990s
 - Widespread use today → If the color space of an image is unknown, sRGB is the assumed default
 - Not just a 3x3 linear transformation from (R_C, G_C, B_C) or (R_N, G_N, B_N) ; has a gamma correction too (nonlinear)

Some RGB Color Spaces (Wikipedia)

Color space	Reference Standard	Year	White point	Primaries						Display	Transfer function parameters				
				Red		Green		Blue		gamma	Y	α	β	δ	βδ
				x _R	y _R	x _G	y _G	x _B	y _B	EOTF		a + 1	K ₀ /φ = E _t	φ	K ₀
NTSC-J	Based on NTSC(M)	1987	D93	0.63	0.34	0.31	0.595	0.155	0.07	2.5	$\frac{20}{9}$	1.11150.005740.0228			
NTSC, MUSE	SMPTE RP 145 (C), 170M, 240M	1987	0.625			0.28				1.8					
Apple RGB	(Apple Computer)			0.64	0.33	0.29	0.60	0.15	0.06	2.8	$\frac{14}{5}$	1.0550.003130812.920.04045			
PAL / SECAM	EBU 3213-E, BT.470/601 (B/G)	1970	0.30			2.2				$\frac{12}{5}$					
sRGB	IEC 61966-2-1	1996, 1999				2.4				$\frac{20}{9}$					
scRGB	IEC 61966-2-2	2003	0.21			0.71	0.14	0.08	2.2	$\frac{563}{256}$	1.0990.0044.50.018				
HDTV	ITU-R BT.709	1999							2.8						
Adobe RGB	(Adobe)	1998	0.67			0.21	0.71	0.14	0.08	2.5	$\frac{11}{5}$	1.160.0088569.0330.08			
M.A.C.	ITU-R BO.650-2 ^[4]	1985		2.2											
NTSC-FCC	ITU-R BT.470/601 (M)	1953		0.68	0.32					0.265	0.69				
PAL-M	ITU-R BT.470-6 ^[5]	1972	~2.2			$\frac{12}{5}$									
eciRGB	ISO 22028-4	2008, 2012	D50			2.4		1.09930.0180544.50.081243							
DCI-P3	SMPTE RP 431-2	2011	6300K	0.708	0.292	0.170	0.797	0.131	0.046	2.2	$\frac{563}{256}$	1.0990.00185.50.099			
Display P3	SMPTE EG 432-1	2010	D65							2.222	$\frac{20}{9}$				
UHDTV	ITU-R BT.2020, BT.2100	2012, 2016		D50	0.7347	0.2653	0.1596	0.8404	0.0366	0.0001	1.8	$\frac{9}{5}$	1	0.001953125	16
Wide Gamut	(Adobe)		E				1	0	0.2738	0.7174	0.1666	0.0089			
RIMM	ISO 22028-3	2006, 2012		0.7347	0.2653	0.1596			0.8404	0.0366	0.0001	1.8	$\frac{9}{5}$	1	
ProPhoto (ROMM)	ISO 22028-2	2006, 2013	E			1	0	0.2738	0.7174	0.1666	0.0089				
CIE RGB	CIE 1931 color space	1931		1	0			0	1	0	0		1		
CIE XYZ															

Can't you display all colors with RGB primaries?

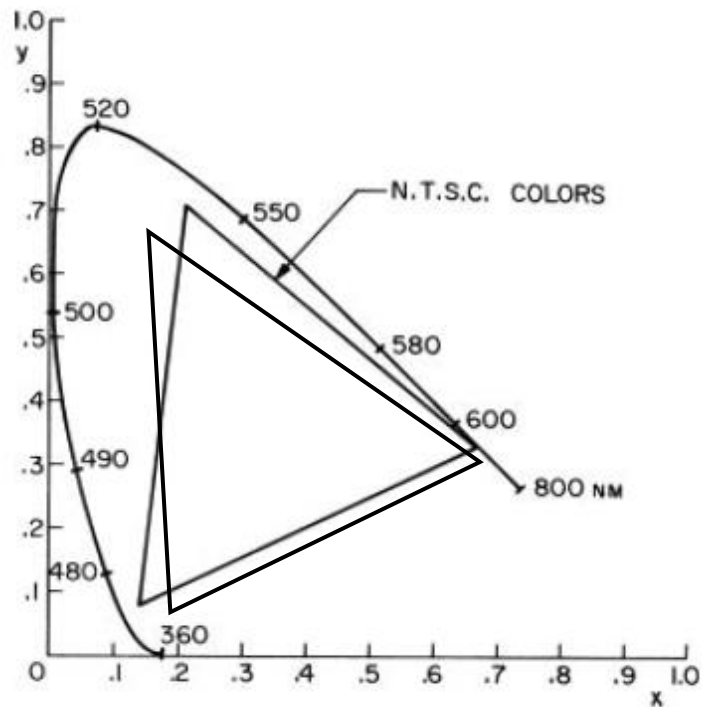
1. What do you mean by RGB?
2. Answer: No, you can't



From: <https://www.rtings.com/laptop/reviews/apple/macbook-air-13-m1-2020>

Which gamut is better?

How can we answer this question?
Measure the areas??



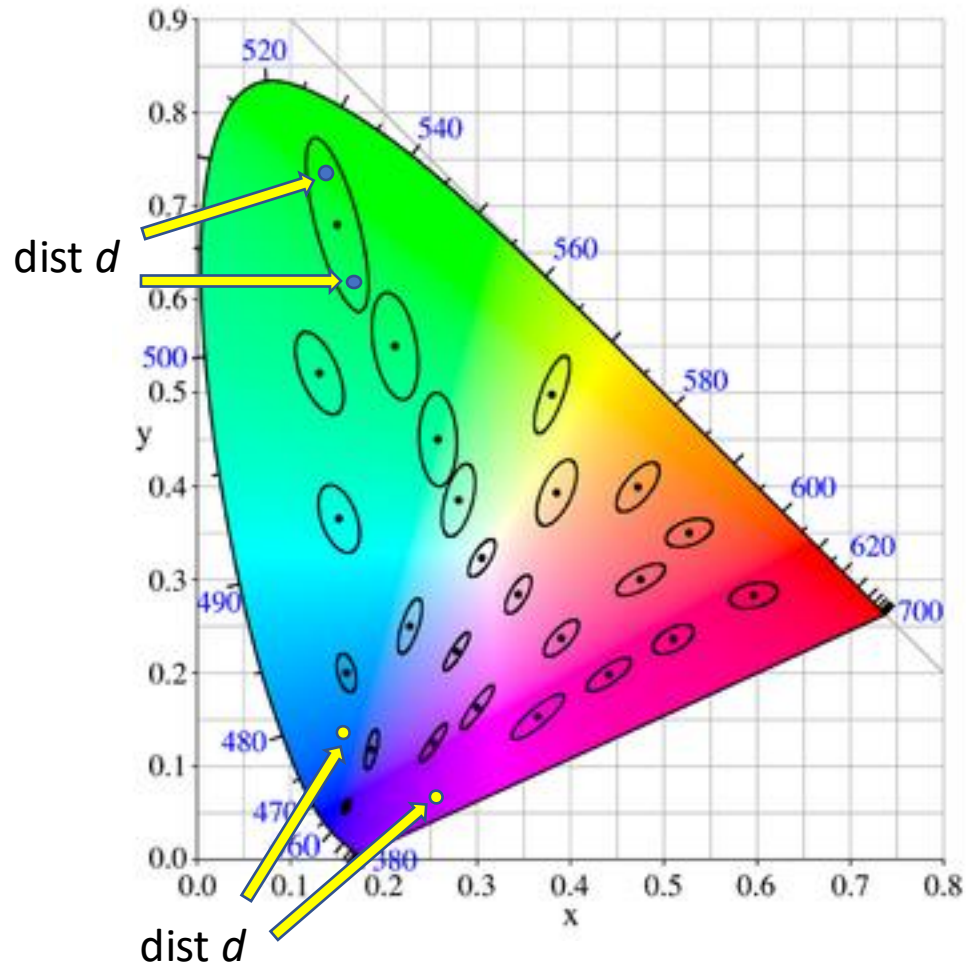
How different are two colors?
Measure the Euclidean distance between them?

This space is not perceptually uniform

Uniform Color Spaces

1. Comparison of color gamuts
 - Want perceptually distinct colors
2. Specification of color tolerances: When a manufacturer orders a colored part from a supplier, they must specify a color tolerance.
 - Based on human perception
3. Specification of color codes: Using colors to code data, sometimes need the colors to be as distinct as possible:
 - Cockpit display for airplane, or display for air traffic control
 - Different wires in a cable
4. Pseudocolor sequences for maps:
 - Used in medical imaging, geophysics, astronomy, etc.
 - A uniform color space can be used to create perceptually equal steps in a sequence of colors

MacAdam Ellipses

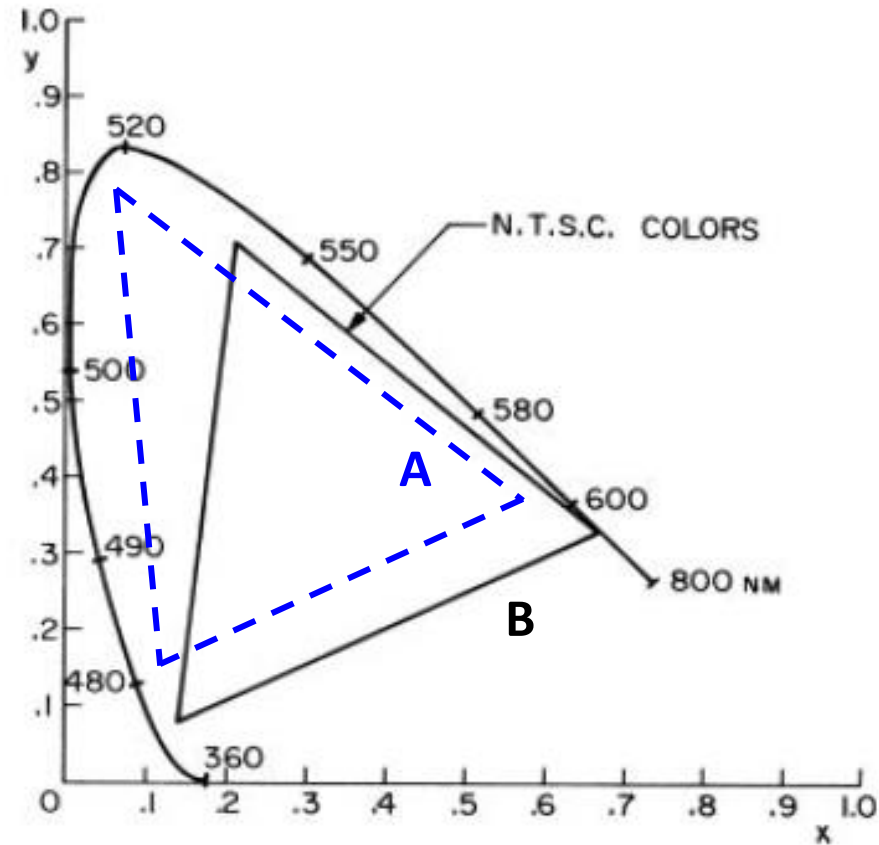
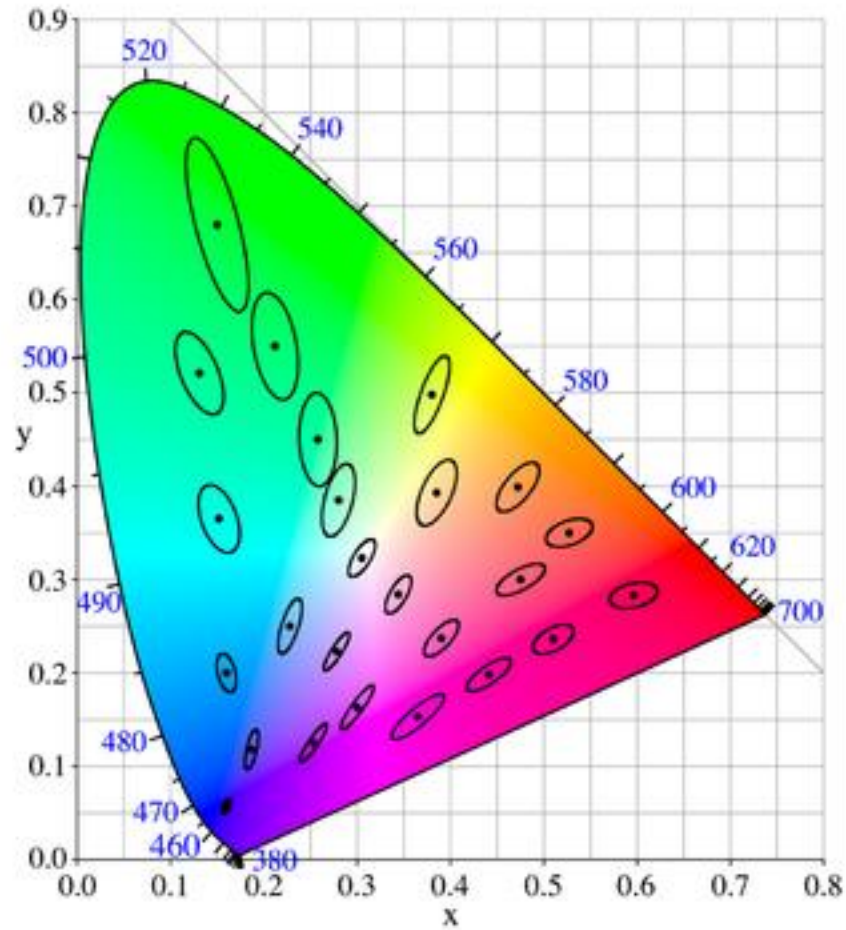


- A MacAdam ellipsis is a region on a chromaticity diagram that contains all colors which are indistinguishable to the average human eye from the color at the center of the ellipse
- Trained observer does color matching experiment, repeatedly, for 25 different colors
- Matches are fit with an ellipse
- Shown here at 10X size
- We'd like to have some space where these are circles! And the *same size*.

Question

Suppose the area of the blue-dashed triangle (A) is equal to the area of the black triangle (B)

Which would likely correspond to a larger set of perceptually distinct colors?

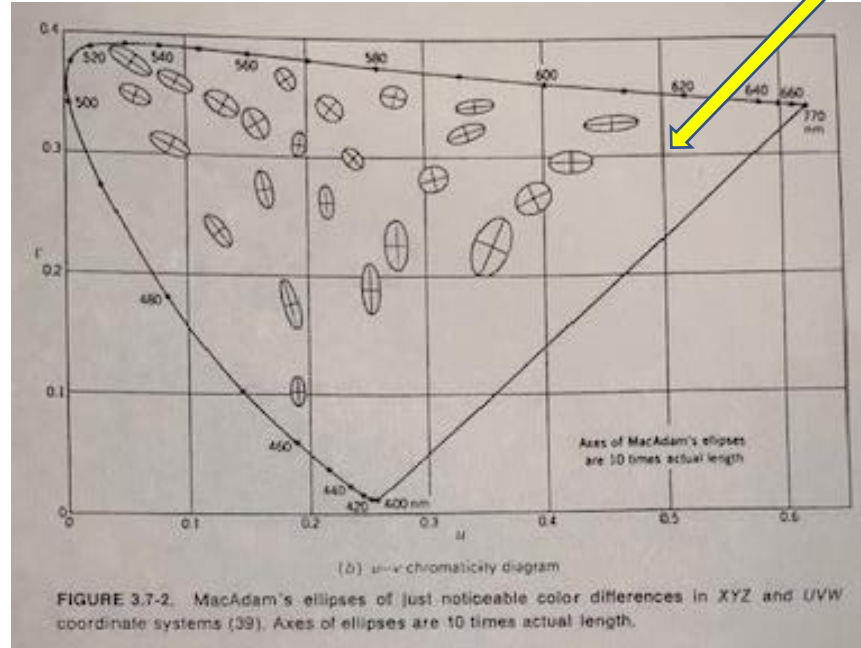


Attempts at perceptual uniformity

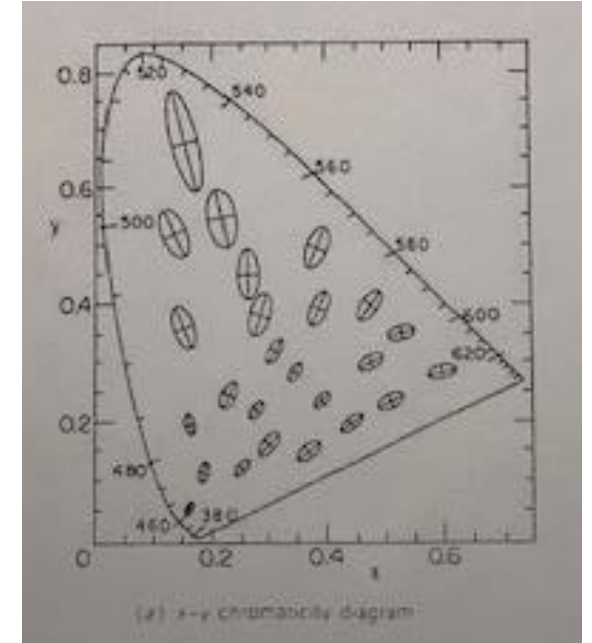
1. UVW color space in 1960, achieved some uniformity in chromaticity

$$\begin{bmatrix} U \\ V \\ W \end{bmatrix} = [3 \times 3] \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

2. Extended in 1964 to U'V'W'
 - Attempting to be uniform in brightness also



closer to being circles,
and being same size



3. Linear transformations are not sufficient

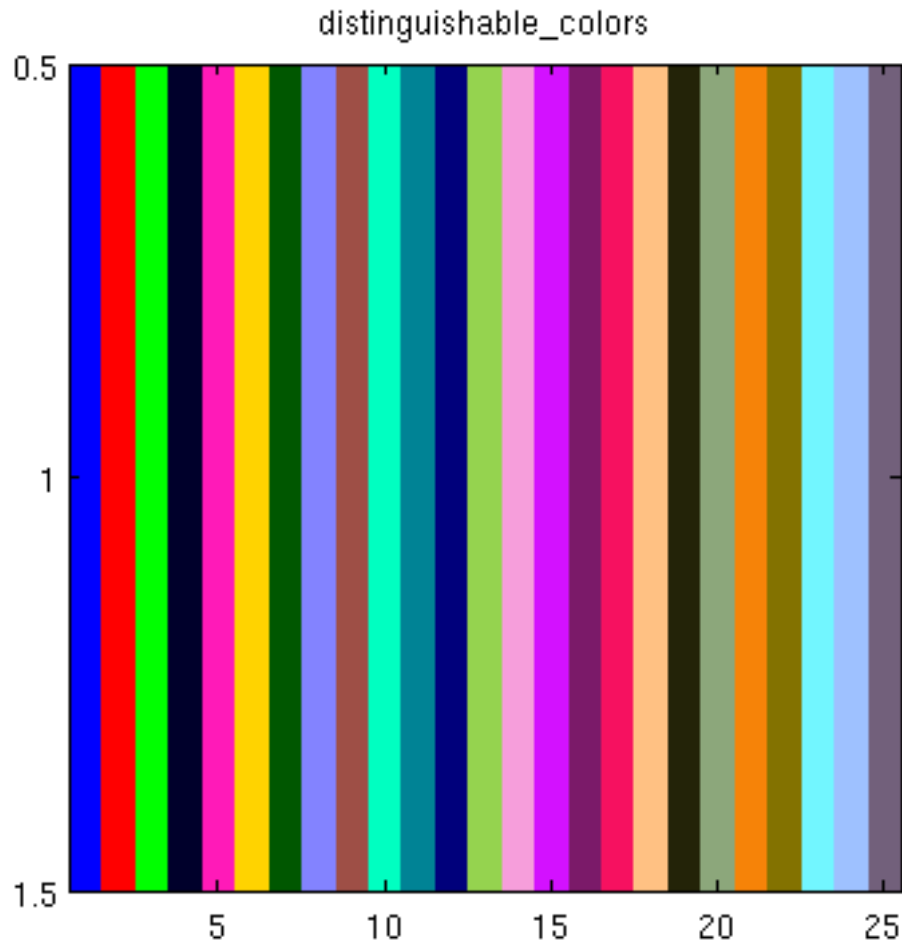
u-v

x-y

- UVW & U'V'W' replaced by CIE Lab and CIE L*a*b*
- Can use Euclidean distance to measure perceptual distance in CIE L*a*b*

Example application: Creating distinguishable colors for lines on a plot

```
function colors = distinguishable_colors(n_colors,bg,func)
```



Choose points in Lab space that are maximally far apart

Tim Holy (2022). Generate maximally perceptually-distinct colors (<https://www.mathworks.com/matlabcentral/fileexchange/29702-generate-maximally-perceptually-distinct-colors>), MATLAB Central File Exchange.

Categories of color coordinate systems

1. CIE RGB: (R_C, G_C, B_C) The monochromatic spectral primaries, used in the 1920's for the original color matching experiments
2. CIE XYZ: Used for color specification, all colors are in the positive quadrant
 - Reference system
3. For color display: RGB models
 - NTSC RGB
 - sRGB
4. Aiming at perceptual uniformity
 - UVW, $U^*V^*W^*$
 - Lab, $L^*a^*b^*$
5. Color Printing
 - CMY, CMYK
6. Color Compression & Transmission
 - YIQ, YUV, YCrCb
7. Color Image Processing?
 - Hue, Saturation, Intensity (HSI)
 - Hue, Saturation, Value (HSV)
 - Hue, Saturation, Lightness (HSL)

Subtractive color matching

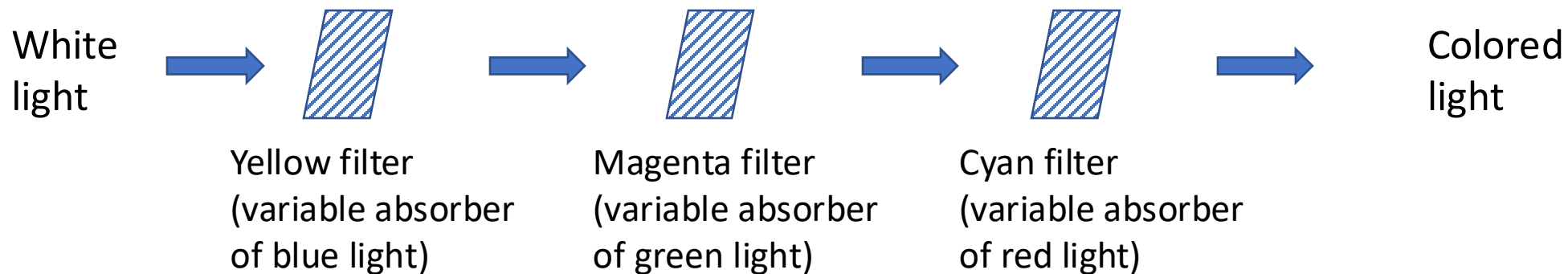
1. So far: Additive color matching:

- Start with zero intensity of 3 primaries = Black
- Progressively add more → Eventually get White

2. Subtractive color matching:

- Start with white light, nothing filtered out
- Progressively filter out more → Eventually get Black

$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$



Color Printing Systems: CMY & CMYK

$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

In practice, need to have a separate black ink (K)

- Reduces need for large amounts of colored ink (cost)
- Impurities in the ink
- Thickness of build-up
- Reflections off surface of ink (instead of passing through all layers)

Converting from CMY to CMYK:

$$K = \text{Black Ink}$$

When R=G=B=0 in the RGB system, this corresponds to pure black:

- C=M=Y=1 in the CMY system
- K=1 and $C_{CMYK} = M_{CMYK} = Y_{CMYK} = 0$ in the CMYK system

Otherwise:

$$K = \min(C, M, Y) \quad C_{CMYK} = \frac{C - K}{1 - K}$$

$$M_{CMYK} = \frac{M - K}{1 - K} \quad Y_{CMYK} = \frac{Y - K}{1 - K}$$

Systems for Analog Color Transmission: YIQ, YUV

- Analog B&W TV, 1941, NTSC standard settled, first commercial broadcast in US
- 1946: CBS gave FCC first demo of their color TV
 - 72 frames/sec with R,G,B frame interleaving
 - Incompatible with B&W systems
- RCA flooded market with B&W sets to slow the potential launch of CBS color
- 1950: FCC approved CBS color TV system
 - It died
- 1953: FCC approved RCA color TV system
 - B&W compatible

$$\begin{bmatrix} Y \\ I \\ Q \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ 0.596 & -0.274 & -0.322 \\ 0.211 & -0.523 & 0.312 \end{bmatrix} \begin{bmatrix} R_N \\ G_N \\ B_N \end{bmatrix}$$

Y = luminance or luma

I,Q called inphase and quadrature components

- chromaticity information
- less important, could be sent in smaller bandwidth
- Why is it less important?

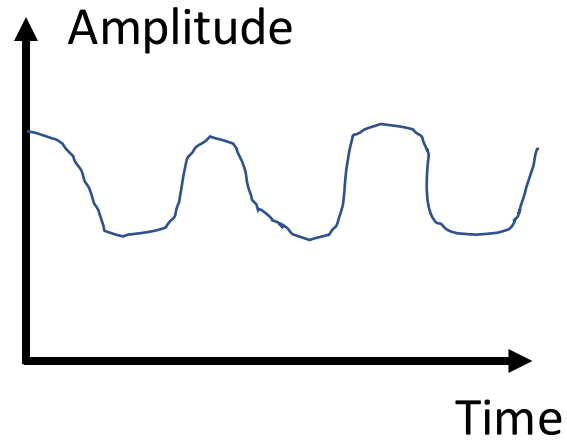
Notice: Sum of row 1 = 1

Sum of row 2 = 0

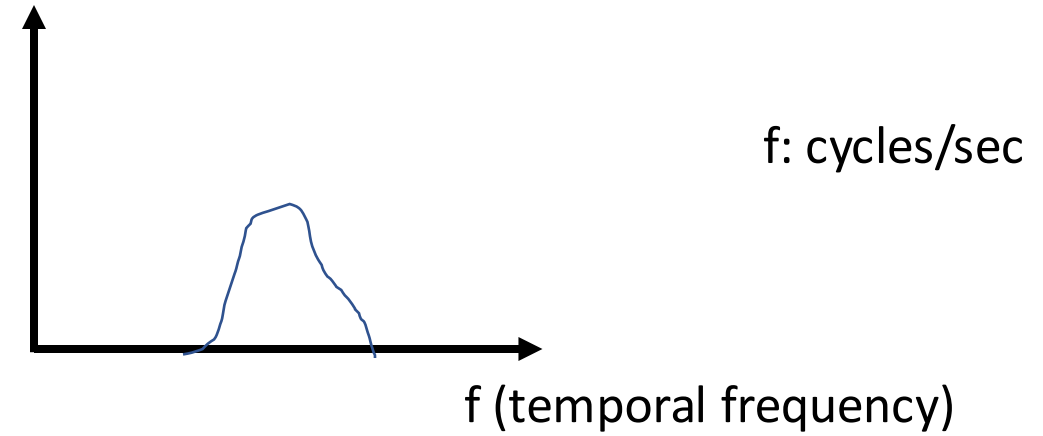
Sum of row 3 = 0

So: when R=G=B then I=Q=0, and Y=R=G=B

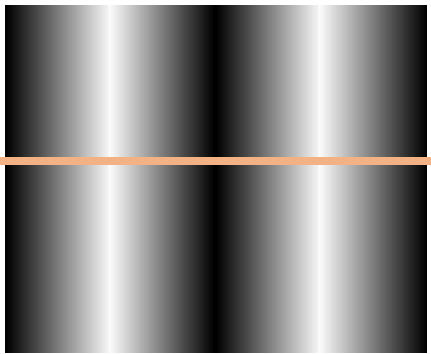
Temporal Frequency & Spatial Frequency



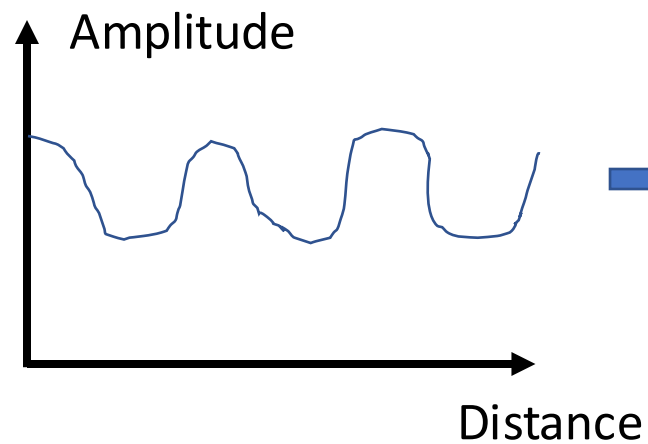
FT



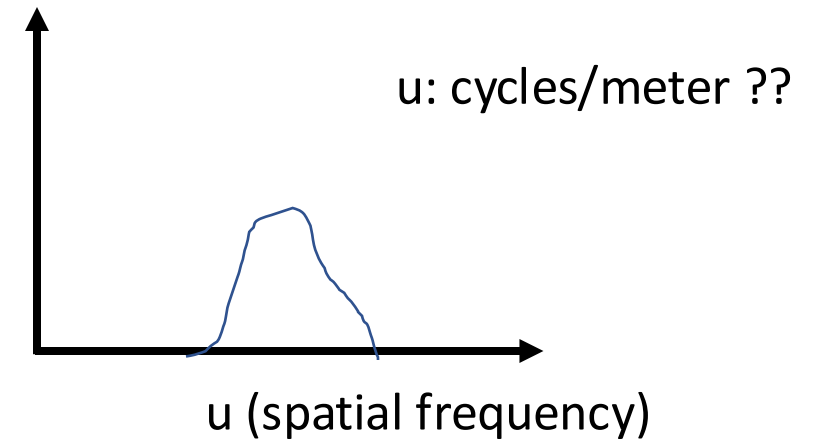
Image



Take slice
through
image

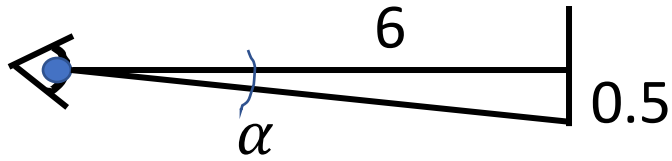


FT



Different Possible Units for Spatial Frequency

- Analogous to temporal frequency (cycles per second), spatial frequency can be measured in cycles per unit distance
- But we're also concerned how it looks to the eye:

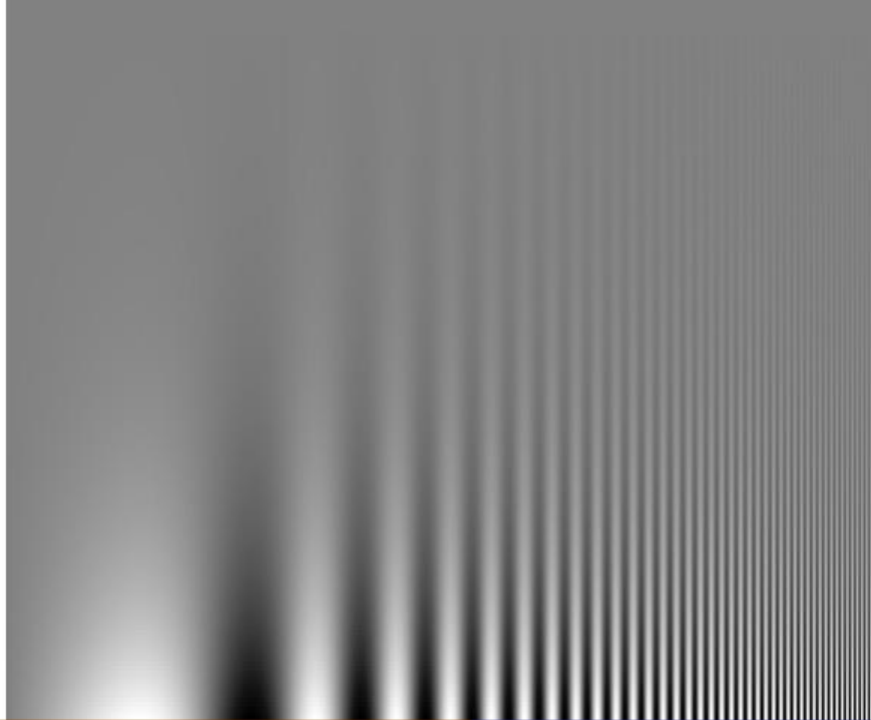


- $\tan \alpha = 0.5/6 = 1/12$
- $\alpha \approx 4.76$ degrees

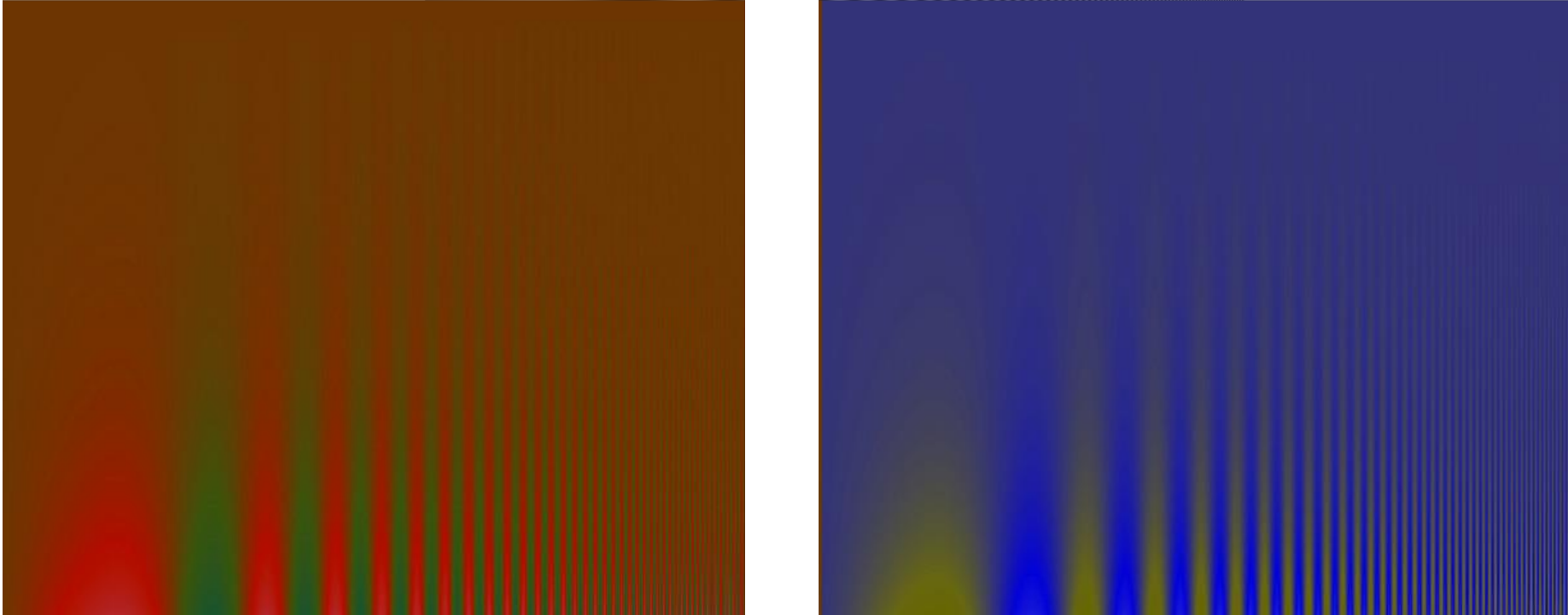
- Suppose half screen width = 3600 pixels
- Suppose some pattern has 1 cycle per 320 pixels

$$\frac{3600 \text{ pixels}}{4.76 \text{ degrees}} \times \frac{1 \text{ cycle}}{320 \text{ pixels}} = 2.36 \text{ cycles per degree of visual angle} = 2.36 \text{ cpd}$$

Spatial Frequency Sensitivity



- Human visual system is not sensitive to high spatial frequencies
- And is even less sensitive to high spatial frequencies when they are in color



Color Systems for Video Transmission: YIQ & YUV (analog), YCbCr (digital)

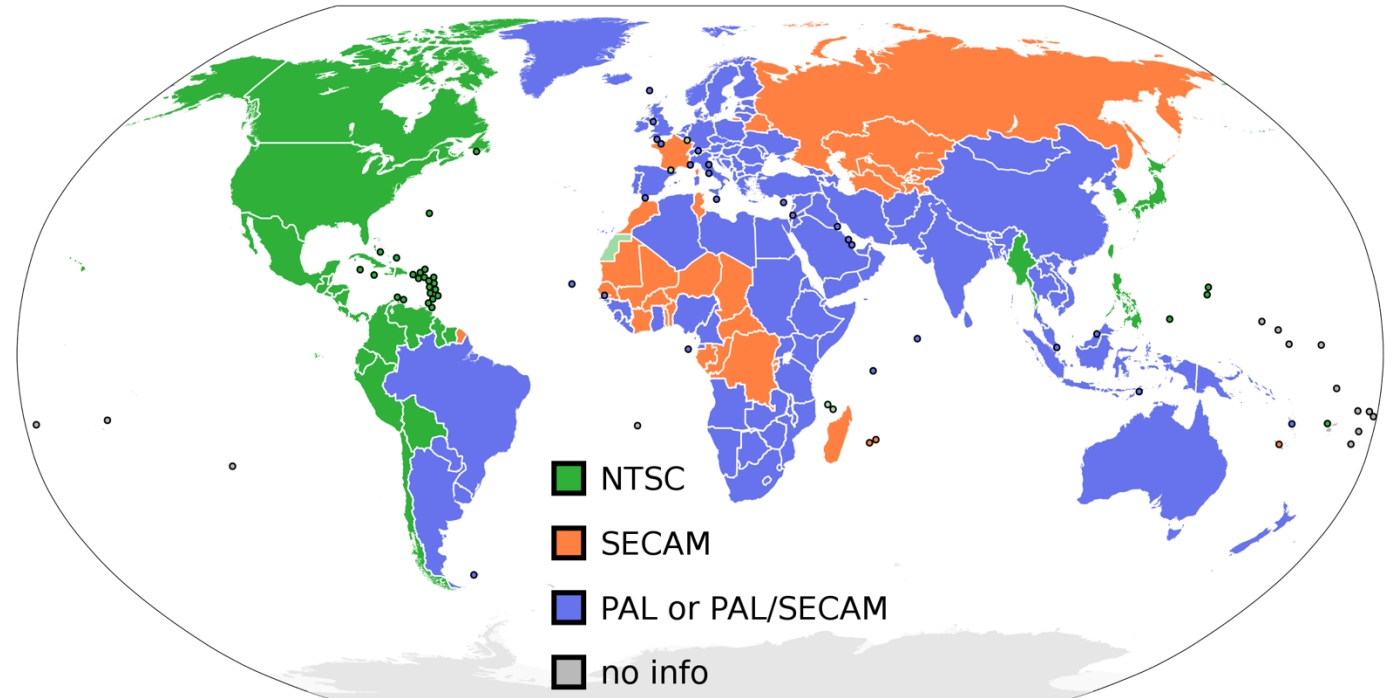
- Elsewhere: PAL, SECAM → YUV
- *Sequential Color with Memory*

$$\begin{bmatrix} Y \\ U \\ V \end{bmatrix} = [3 \times 3] \begin{bmatrix} R_N \\ G_N \\ B_N \end{bmatrix}$$

- SECAM = “*Système Essentiellement Contre les Américains*”

NTSC = Never Twice the Same Color

- Digital storage application: JPEG & MPEG Compression
- YCbCr: similar idea, chromaticity planes get downsampled



These systems:

- have one color plane for luminance/intensity
- have two color planes for chromaticity information
- chromaticity given less bandwidth or fewer bits